Patents, Innovation, and Market Entry

by

Dominik Jurek
Haas School of Business, UC-Berkeley

CES 23-45 **September 2023**

The research program of the Center for Economic Studies (CES) produces a wide range of economic analyses to improve the statistical programs of the U.S. Census Bureau. Many of these analyses take the form of CES research papers. The papers have not undergone the review accorded Census Bureau publications and no endorsement should be inferred. Any opinions and conclusions expressed herein are those of the author(s) and do not represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. Republication in whole or part must be cleared with the authors.

To obtain information about the series, see www.census.gov/ces or contact Christopher Goetz, Editor, Discussion Papers, U.S. Census Bureau, Center for Economic Studies, 4600 Silver Hill Road, Washington, DC 20233, CES.Working.Papers@census.gov. To subscribe to the series, please click here.

Abstract

Do patents facilitate market entry and job creation? Using a 2014 Supreme Court decision that limited patent eligibility and natural language processing methods to identify invalid patents, I find that large treated firms reduce job creation and create fewer new establishments in response, with no effect on new firm entry. Moreover, companies shift toward innovation aimed at improving existing products consistent with the view that patents incentivize creative destruction.

Keyword*:* intellectual property rights, creative destruction, entry, job creation,
Alice decision, natural language processing

---

[*] Any views expressed are those of the authors and not those of the U.S. Census Bureau. The Census Bureau's Disclosure Review Board and Disclosure Avoidance Officers have reviewed this information product for unauthorized disclosure of confidential information and have approved the disclosure avoidance practices applied to this release. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Number 2669. (CBDRB-FY22-P2669-R10037)

1. Introduction

Do patents facilitate market entry of new ideas by protecting rents on novel technologies? Or do they stifle innovation by allowing incumbents to maintain excessive legal monopolies? Influential research suggests that patents facilitate the market entry of new ideas by granting innovators property rights over their inventions and incentivizing costly, risky investments in innovation (e.g., Schumpeter 1942, Arrow 1962, Nordhaus 1969). In this way, strong patent protection can spur start-ups and incumbents to introduce new technologies to the market, accelerating economic growth. However, research also suggests that poorly designed patent systems that grant excessively broad property rights can stifle innovation and new entrants (e.g., Boldrin and Levine 2013). For example, incumbent firms can exploit such systems by acquiring patents that block new firms from entering the market, allowing incumbents to extract rents from their existing products rather than developing new technologies to compete with other firms. From this perspective, patents can decrease entry, increase markups for incumbents, and reduce business dynamism and growth (Akcigit and Ates 2019).

In this paper, I contribute to research on how patents affect innovation by evaluating the effect of a recent court decision that limited the patent eligibility of software patents. The 2014 *Alice Corp. v. CLS Bank International* Supreme Court decision[1] (*Alice* decision) weakened the enforceability of existing software patents and limited the patentability of new software-related innovations. I assess the impact of the *Alice* decision on job creation, firm entry and exit, and innovation efforts.

Identifying patents that are affected by *Alice* is challenging because few patents are litigated and software patents fit into multiple broad patent classes. Instead, I introduce a novel approach that directly uses the patent claim language. I use texts of patent applications that were rejected due to the *Alice* decision to train a binary natural language processing (NLP) classification algorithm and identify claim language that is not eligible after *Alice* anymore. The advantage of this method is that I can predict for a large set of patents whether they are invalid and identify treated firms by measuring the share of invalid patents in patent portfolios before the *Alice* shock.

To measure innovation and growth-relevant outcomes, I leverage restricted-use U.S. Census microdata on establishment entry and job creation and patent data on innovation quality

[1] 573 U.S. 208 (2014)

and direction. I estimate a difference-in-differences (DiD) model and address two sets of questions that measure the effect of patentability on innovation and growth: First, how do establishment entry, job creation and destruction, and innovation quality and direction change for incumbents if they receive fewer patents for their inventions? Second, which types of firms are most affected by the *Alice* shock to patentability?

I find negative treatment effects of 5.5% for job creation and 3.4% for new establishment entry, but a positive treatment effect of 2.4% for markups. Growth theory as in Peters (2020) interprets this result as incumbents reducing the exploration and market entry of new technologies and becoming instead more focused on the improvement of their current products. The shift toward own innovation is also reflected in patenting data: while *Alice* has a negative effect on overall issuances, patents become more narrow, less experimental, and focused more on internal rather than external innovation.

Depending on which incumbents shift their innovation strategy, the implications for innovation and growth might be different. If large market leaders reduce exploration, the long-term effects on competition and growth are smaller than if followers limit innovation directed at acquiring new technologies to replace leading firms. I find that the largest firms above the 90th percentile of the patent portfolio and firm employment size distribution drive the results: new establishment creation decreases by 9.8% while overall patent issuances decrease by 10.9%. Markups increase by 2.1%, and patent references to internal knowledge increase relative to overall patent citations. Overall, the *Alice* shock reduces the patentability of innovations, and large incumbents substitute new technology exploration with product improvements. Thus, the results are consistent with patents incentivizing incumbents to innovate by entering into new technologies, which in turn promotes creative destruction and growth.

My results are robust to variations in the regression specification and sample selection. I implement several alternative definitions for treated and control firms, estimate economic outcomes with different variables, and run various model specifications. Furthermore, my approach fulfills all assumptions for the causal identification of treatment effects. For the difference-in-differences setting to be valid, the key assumption is that changes in outcome variables are different for the treated group only due to *Alice*. The *Alice* decision is the most significant change in software patentability since 1998, and no patent regime change affected the control group in the sample period. Thus, only the treated group experiences a consistent shock to patentability. Another concern might be unobserved industry shocks

affecting the treated firms. I find that *Alice*-treated patents are concentrated among large firms in computer-related service industries, but my main results are robust to the inclusion of sector-year fixed effects and controlling for size-related covariates, limiting concerns that the treatment effects are driven by other channels than *Alice* (Goldsmith-Pinkham et al. 2020). Econometrically, the treatment variable is uncorrelated with covariates that are related to changes in outcomes, and graphical analysis of event study outcomes does not reject the parallel trend assumption. Thus, there is no economic or statistical reason why the exclusion restriction should be violated.

This paper contributes to the literature in three ways. First, I show consistent microdata-based evidence for Schumpeterian growth theory with heterogeneous innovations (Akcigit and Kerr 2018, Garcia-Macia et al. 2019, Klenow and Li 2021). Large firms reduce the exploration of new technologies in favor of improving existing products after the *Alice* shock. This fits into the framework introduced by Peters (2020). The core idea is that firms improve their current products to gain market power and increase markups, but also engage in creative destruction to acquire new technologies. My results are consistent with patents reducing expansion costs for incumbents. That is, without the ability to receive patents for their innovations, incumbent firms face higher costs of breaking into new product markets and thus increase own innovation of their current products relative to creative destruction. I show that this is most important for large firms. Since firms substitute innovation types rather than limit innovation efforts, the overall welfare effect is relatively small.[2]

Second, I introduce a novel identification approach combining machine learning, natural language processing, and patent data. Recent literature has used firm and patent texts to measure the similarity of companies and technologies (Hoberg and Phillips 2016, Frésard et al. 2020, Kelly et al. 2021) and group firms and patents based on text analysis concepts such as tone, sentiment, or readability (Loughran and Mcdonald 2011, Ke et al. 2019, Kong et al. 2020). My approach is the first application of NLP-based patent classification to identify causal treatment effects in response to a patent regime change. Moreover, my method provides new insights into the importance of software and business method patents for economic growth (see, e.g., Bessen and Hunt 2007, Webb et al. 2018, Lerner et al. 2021). The role of software patents is best understood in the context of why some low-citation patents

[2] In appendix D, I calibrate a simple model based on Peters (2020) with the estimated treatment effects and find a decrease in welfare of ca. 1.3% for the industries most affected by *Alice*.

have a high value for patent holders. The number of citations a patent receives is generally considered to be a proxy for the value of the innovation (Harhoff et al. 1999, Hall et al. 2005). However, Abrams et al. (2013) find that some high-value patents receive few citations and model low-citation patents as additional protection against infringement for other valuable inventions. Consistent with this idea, I provide direct empirical evidence that the additional protection from low-citation patents is important for new establishment creation. Thus, software patents, which are generally considered to be low-value patents (Hall and MacGarvie 2010), are economically important for their value in protecting innovations.

Finally, I provide empirical support for a positive causal link between patenting and entry with my core insight that large firms protect the market entry of new technologies with patents. This is an important new perspective on the longstanding question of how patentability incentivizes innovation: early theoretical literature provides no clear answer to whether a broad or narrow patenting regime is optimal (see, e.g., Gilbert and Shapiro 1990, Klemperer 1990, Gallini 1992). Empirical evidence on the question is scarce with some studies suggesting that broad patent rights increase innovation costs and limit entry (Cockburn and MacGarvie 2011, Hall et al. 2021), while other studies find little effect of patentability on innovation and follow-on inventions (Moser 2011, Sampat and Williams 2019). On the other hand, studies such as Gans and Stern (2003), Aghion et al. (2014), and Aghion et al. (2015) stress the importance of strong patent protection to incentivize innovation investment.[3] My empirical results support a more differentiated view of patents: patentability incentivizes innovation that is linked to new establishment entry and job creation for large firms but has no effect on innovation by small firms. A more narrow patent regime leads large firms to focus on a different type of innovation (own innovation) rather than reduce innovation efforts. Thus, patents are positively related to entry and job creation through the exploration of new technologies.

[3] My research is closest related to Cockburn and MacGarvie (2011) who ask whether patents on relevant technologies keep firms out of software product markets. In a sample from 1990 to 2004, they find that a 10% increase in the number of patents relevant to a market reduces the rate of entry by 3%–8%, which intensifies following the expansion of the patentability of software in the mid-1990s. I directly build and improve upon this result in two ways: first, my analysis can capture how large firms can change their innovation strategy by focusing on own innovation. Second, unlike previous studies that look at expansions of what can be patented (Cockburn and MacGarvie 2009, Noel and Schankerman 2013), my study focuses on a regime change in the opposite direction. This distinction is economically important because the response to patenting regime changes might not be symmetric.

2. Methodology and data

I exploit a recent Supreme Court decision that limited the patent eligibility of software patents as a natural experiment to identify the causal effect of patentability on innovation-relevant outcomes. I develop a novel NLP classification method on patent texts to identify technologies that became patent ineligible following the Supreme Court decision. I define treated firms based on their share of affected patents and estimate treatment effects with a difference-in-differences (DiD) approach. To measure growth and innovation-relevant outcomes, I use restricted-use U.S. Census data on establishment counts and employment at the firm level and patent quality measures from the innovation literature.

2.1. Identification

To identify the causal impact of patentability on innovation, I need to find an exogenous shock that changed the ability to receive patents for inventions. Software-related patents are a good starting point for this: software concepts are difficult to describe in patent claims leading to uncertainty about the right scope of patent-eligible subject matter. On the one hand, the incentive to innovate needs to be preserved while avoiding on the other hand 'patent thickets' that stifle competition and limit innovation by other inventors (Stroud and Kim 2017). As a consequence, Supreme Court decisions led to several regime shifts over the decades on what is and is not patent-eligible.

Starting with the *State Street* decision[4] in 1998, a relatively broad interpretation of patent eligibility was adopted. In the following years, overly broad software patents were linked to the rise of patent trolls and nonpracticing entities; a type of patent holder that generates income by claiming license fees and litigating against genuine innovators (see, e.g. Appel et al. 2019, Cohen et al. 2019, Lee et al. 2019, Lemley and Feldman 2016). This broad patentability regime came to an end with the *Alice* decision: on June 19, 2014, the U.S. Supreme Court affirmed the judgment of a lower court that the patent claims directed at a method for mitigating settlement risk were ineligible because the claims just implemented an abstract idea (Ren and Duprez 2019). This decision, as many commentators were quick to point out, increased uncertainty around the patent eligibility of software and business method claims and limited their scope, reversing the prior broad patentability regime for software innovations (see, e.g. DiNizo 2018, Tran 2015, 2016, Daily 2017, Stroud and Kim 2017, Chien

[4] *State St. Bank & Trust Co. v. Signature Fin. Group, Inc.*, 149 F.3d 1368 (Fed. Cir. 1998).

2016, Craig 2017). Recent empirical research has confirmed the impact of *Alice* on patenting and litigation: Chien and Wu (2018), Kesan and Wang (2020), and Toole and Pairolero (2020) all find that rejection rates and application abandonments among software and business method application have spiked after *Alice*, with some rejection rates for business methods growing from 25% to 81% in the month after *Alice*.

Overall, the *Alice* decision is one of the most important recent regime shifts for the patentability of certain types of innovations. For individual firms, this means an exogenous shock to the ability to file for patent protection of new inventions and enforce existing software patents.

To use this regime shift to identify the causal effect of patentability, I need to classify patents based on their exposure to the *Alice* decision. This is challenging since not one particular type of patent or patent class was invalidated, rather a certain type of language describing innovations was ruled to be patent-ineligible. Prior research uses United States Patent Classification (USPC) patent class 705 to identify the most treated patents (see, e.g., Wagner and Cockburn 2010, Contigiani 2020). I can confirm that this class is indeed the most affected USPC class by *Alice*, but USPC classes are not useful going forward since they were discontinued by the United States Patent and Trademark Office (USPTO) in 2013. Using USPC classes thus does not allow us to validate the empirical approach and identify treated patents after the court decision. The Cooperative Patent Classification (CPC) system, which replaced the USPC system, has several drawbacks for treatment identification as well: there is no direct concordance between USPC and CPC, and CPC groups are much broader than USPC classes, thus covering more technologies than the patent-ineligible inventions after *Alice*. Another approach to identify treated patents would be to directly observe litigated patents after *Alice* (Galasso and Schankerman 2015). While this certainly allows us to define which patents are not eligible under *Alice*, only about 1.5% of patents are ever litigated (Lemley and Shapiro 2005).

To allow for a consistent measure of treated patents across the entire sample, I develop and train an NLP model to identify claim language that is invalid under *Alice* and classify existing patents. I use applications that were rejected due to *Alice* in the Office Action Research Data from Lu et al. (2017) as a sample of text formulations that are not eligible after the decision. I select issued patents from the same filing years and patent classes as the rejected applications to obtain control claims with eligible language. I extract the rejected

independent claim texts from the application publications and collect from PatentsView the published independent claims of issued control patents. I then train a support vector machine on a TF-IDF matrix of the rejected and control claim texts, which allows me to predict for a given claim text whether it would be rejected under *Alice* or not. I use this NLP model to classify all independent claims of issued patents since 1990 in PatentsView in CPC groups related to the training data. In total, this results in 3,359,812 classified independent claims in 1,062,897 patents.[5] Appendix E provides a detailed description of the NLP methodology, analysis of the language features of *Alice*-affected claim texts, and validation tests for the quality of the patent classification. Overall, all performance metrics point toward a good classification model with high precision (91.7%), high recall (94.1%), and more than 85% Matthews correlation coefficient (MCC), meaning a very high correlation between the true and predicted labels. I can thus predict for each patent in related patent classes if it is affected by *Alice* or not. I define a patent as treated if my NLP method predicts that its first claim is affected by *Alice*. The first claim is generally the broadest claim of a patent and thus the most relevant text when defining which technologies are covered by the patent (see, e.g., Kuhn and Thompson 2019).[6]

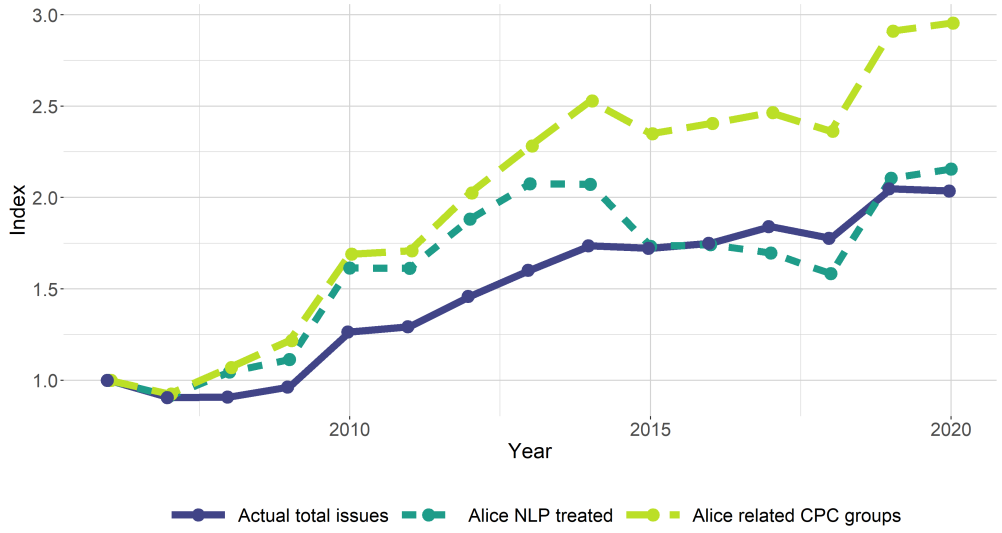
To validate my approach, figure 1 shows the indices of treated patent issuances defined by my NLP method (*Alice* NLP treated), patents in the CPC groups closely related to *Alice* (*Alice*-related CPC groups), and all utility patent issuances in a given year (Actual total issues). Following 2014, there was a clear downward shock for both, the *Alice*-related CPC groups as well as my *Alice* NLP-treated patents, while actual total issues did not decrease. My NLP classification method identifies treated patents better than the affected CPC groups alone, which are broad and cover more technologies than are treated by *Alice*: the *Alice*-related CPC groups saw a decline in patent issuance from 2014 to 2015 of 7.1%, while *Alice* NLP-treated patent issuances declined by more than 16.3%.

Several other observations are interesting in this graphic: first, software-related patent issuances in *Alice*-related CPC groups are rising faster than other types of patents before

[5] CPC classifications were retroactively assigned to already issued patents, thus I can identify the CPC groups closest related to the USPC classes with the most *Alice*-related rejections in my training data (705, 463, 434, and 702). I restrict the five most treated CPC groups, which are CPC groups A63F (Video games), G07F (Coin-freed or like apparatus), G06F (Digital data processing), H04L (Transmission of digital information), and G06Q (Data processing systems).

[6] The following analysis is robust to alternative definitions of treatment, such as requiring at least one independent claim to be treated.

Figure 1 Normalized patenting index 2006 to 2020.



Note. Count of issued utility patents by year, normalized to 2006 levels. The three patent indices are the actual count of all issued utility patents, the count of patents that are classified by the NLP algorithm as treated under *Alice*, and the count of issued patents in the five CPC groups closely related to *Alice* (A63F (Video games), G07F (Coin-freed or like apparatus), G06F (Digital data processing), H04L (Transmission of digital information), and G06Q (Data processing systems)). Patent data are sourced from PatentsView.

Alice, consistent with the evidence in Lerner et al. (2021) and Webb et al. (2018). A jump in annual issuances in 2010 is followed by a flattening of the trend relative to total patent issuances. This is consistent with several rulings by the Supreme Court building up to the *Alice* decision, starting with the *Bilski* case[7] in 2010. The flat trend line after 2010 supports the parallel trend assumption before 2014 for my DiD approach, confirming that my pre-period should be defined as the years between 2010 to 2014. We also see an uptick in software patents after 2019, consistent with Toole and Pairolero (2020) describing that the recent subject matter eligibility decision in *Berkheim v. HP, Inc* in 2018[8] and the 2019 revised subject matter eligibility guidance by the USPTO partially reversed the trend in rejections and decreased uncertainty after *Alice*. Thus, the treatment period should end in 2019.

2.2. Data

To answer my research question, I combine patent and restricted-use Census firm-level data to measure the effect of patentability on growth and innovation-relevant outcomes. All variable definitions are summarized in appendix A.

My main dataset is the redesigned Longitudinal Business Database (LBD) (Chow et al. 2021, Jarmin and Miranda 2002). LBD is a restricted-use dataset provided by the U.S. Census

[7] *Bilski v. Kappos*, 561 U.S. 593 (2010).

[8] *Berkheimer v. HP Inc.*, 881 F.3d 1360 (Fed. Cir. 2018).

Bureau covering private, non-farm business establishments with employees that operate in the United States. One of the main goals of the LBD is to measure the entrance and exit of businesses and yearly employment changes. The data are based on the Business Register and use among other source tax data. LBD thus gives a comprehensive picture of the business activities of even small establishments in the US. I focus on the years 2010 to 2019 in LBD; this is consistent with the pre-period of 2010 to 2014 where the parallel trend assumption for treated patent issuances in figure 1 seems to hold well.[9]

To estimate the effect of *Alice*, I aggregate establishment data in LBD to the firm-year level and add revenue data from Haltiwanger et al. (2019). The revenue data are based on tax receipts in the Standard Statistical Establishment List and the Business Register and are available for ca. 80% of all firms up to 2018. I use the novel patent assignee-firm link from Dreisigmeyer et al. (2018) to match patent data to firms. This bridge builds on prior work by Graham et al. (2018) and uses a triangulation method that combines string matching of company names to assignee names and inventor names to employee data in restricted-use Census data creating a high-quality match of patents to firms. I follow prior literature and define measures for novelty, scope, and quality of innovations on the firm level using data from PatentsView. Appendix A describes in detail the construction of the different measures. I focus on patent grant years rather than application years since *Alice* is primarily an issuance shock.[10]

I define the treatment variable as the share of *Alice*-treated patents relative to all issued patents to the firm between 2000 and 2009. This ensures that the pre-treatment period with several court cases and decisions before *Alice* does not lead to sample selection bias caused by firms selecting into treatment by filing *Alice*-affected patents they expect to be invalidated in the future. Furthermore, the parallel trend assumption can be tested for the years between 2010 and 2014. The only caveat of this definition of treatment is the limitation to incumbents patenting before 2010. In appendix B, I perform industry-level analysis to estimate the effect

[9] My analysis uses the 2019 LBD vintage, which includes two years of observations after the last Economic Census in 2017. Substantial efforts are spent to correctly time the birth and death of establishments between Economic Census years, this is not possible though for the last two years of my sample. However, re-timing is most relevant for smaller incumbents. Large firms are likely to be consecutively surveyed by, e.g., the Corporate Organization Survey (COS), and newly created firms are immediately recognized in the underlying tax data. In any case, controlling for re-timing and re-activation of establishments after some years without employment does not change my results.

[10] The results are consistent if I use application years instead. There is a noticeable pre-trend, though, since treated applications filed before 2014 are rejected after *Alice*.

of *Alice* on new firm creation and destruction and find that there was no impact on new firm entry or exit, thus I focus on incumbent firms in the following.

I use the vintage-consistent four-digit NAICS codes from Fort and Klimek (2018) to define industries. LBD only tracks non-farm businesses, following Kerr and Nanda (2009), I additionally exclude all establishments in agriculture, forestry, fishing (NAICS 11), public administration (NAICS 92), private households (NAICS 814), U.S. Postal Service (NAICS 491), restaurants and food stores (NAICS 722 and 445), hospitals (NAICS 622), education services (NAICS 61), and social services (NAICS 624 and 813). Many finance papers also exclude observations in utilities (NAICS 22) and finance and real estate (NAICS 52 and 53) due to these industries being highly regulated (see, e.g., Gutiérrez and Philippon 2016); there is no immediate reason why these industries should be excluded in my study on intellectual property rights and innovation (the results remain unchanged even if utilities and finance and real estate industries are excluded). I assign firms to the four-digit NAICS industry in which they have the highest level of employment.[11]

2.3. Methodology

I use a difference-in-differences setting to estimate the causal treatment effect of *Alice* on innovation and growth outcomes. I estimate a two-way fixed effects (TWFE) regression model on firm-level LBD data for the years between 2010 and 2019, with the post period starting in 2015.[12] The treatment variable is the share of treated patents among all patents issued to the firm between 2000 and 2009.[13]

[11] It should be noted that Compustat as a data source for publicly listed firms is a poor substitute for the following analysis. Using data from Kogan et al. (2017) to identify patenting firms in Compustat, there are only 1,586 firms with positive patent assignments between 2000 and 2009. Of those firms, ca. 1,100 are in manufacturing sectors, thus outside the industries that are mostly treated by *Alice*. Furthermore, there are 12,100 firm-year observations for patenting firms in Compustat between 2010 and 2019, far smaller than the 197,000 observations using LBD. Even among large firms in LBD with more than 1000 employees, less than half can be found in Compustat and the sample is heavily tilted toward manufacturing firms (3,550 out of 4,280 observations). Thus, with the far smaller sample size and the heavy focus on manufacturing, Compustat data are not equivalent to LBD, especially since innovation- and growth-relevant measures such as establishment entry are not directly observable in Compustat. We could try to weight observations based on how important patents are for the company by, e.g., using the ratio of patent portfolio size relative to sales. However, this would increase the bias toward manufacturing firms since the average manufacturing firm has a patent portfolio-to-sales ratio of 0.35, while firms in service sectors have ratios of 0.09–0.19. Overall, Compustat is a poor alternative to LBD for patenting firms due to the industry focus on manufacturing firms which cannot be corrected by reweighting.

[12] For the following analysis, it should be noted that LBD variables such as employment and establishment count are measured in March for a given year, while the innovation and patenting outcomes are measured at the end of the calendar year. Thus, the first full year for the post-period for LBD variables is 2016.

[13] The results remain unchanged if I define treatment instead as the share of patents in *Alice*-related CPC groups without further classification using the NLP method. Since software-related CPC groups include additional technologies that are not affected by *Alice*, the treatment effects are smaller and less precisely estimated.

The main regression model is the following:

$$Y_{i,t} = \alpha_i + \lambda_t + \delta_{Alice} * d_i * Post + X'_{i,t} * \beta + \epsilon_{i,t} \tag{1}$$

where $Y_{i,t}$ is the outcome variable, d_i is the treated patent share, $Post$ the post-period dummy, and the coefficient δ_{Alice} measures the post-treatment effect. α_i and λ_t are fixed effects for the firm and year. Depending on the specification, firm-level controls, $X'_{i,t}$, are the lagged overall employment of the company, the patent portfolio size, and the firm age, all of which are determined at the same time as the continuous treatment (at the beginning of 2010) and interacted with the post-dummy or year dummies (Goldsmith-Pinkham et al. 2020).

To test the necessary parallel trend assumption for identification, I run an event study model for the years between 2010 and 2019:

$$Y_{i,t} = \alpha_i + \lambda_t + \sum_{q=-4}^{5} \delta_{Alice,q} * d_{i,q} + X'_{i,t} * \beta + \epsilon_{i,t} \tag{2}$$

where $d_{i,q}$ is the treatment variable interacted with a dummy for the respective year q relative to 2014. For the parallel trend assumption not to be rejected, we need to observe insignificant effects in the years before 2014 for the coefficients $\delta_{Alice,q}$ and only significant effects in the post-period. I normalize the coefficient for the year 2013 to zero, the year immediately before *Alice*, i.e., $\delta_{Alice,-1} = 0$.

One challenge for DiD models like these is estimating standard errors (Bertrand et al. 2004). In general, my setting fits well to what Abadie et al. (2017) describe as a design problem for which clustering on treatment level, i.e., on firm level, is appropriate. However, we might also be worried about correlated standard errors within industries and sectors. For robustness, I estimate the models in section 3.4 also with clustered standard errors on the industry level.[14]

[14] We might be concerned that larger clusters increase the bias of standard error estimates. Angrist and Pischke (2009) suggest that clustered standard errors even with less than 30 groups lead to reasonable results (see also Bertrand et al. 2004, Hansen 2007).

3. Results

My study provides three main results: first, *Alice* decreases new establishment entry by incumbent multi-unit firms but leaves new venture creation and single-unit entry unchanged. Thus, patentability has little effect on the extensive margin of firm creation but is relevant for incumbent firm growth. Second, companies receive fewer, narrower patents, and citation measures show more usage of internal knowledge after *Alice*. Third, the largest, most innovative firms drive the results. Firms in the top decile in terms of employment and patent portfolio size reduce new establishment and job creation following the limited patentability of inventions and cite more internal knowledge. Overall, the three results are consistent with patentability incentivizing creative destruction for large incumbents.

Following the insight from appendix B that new firm creation is not affected by *Alice* and multi-unit establishment entry decreases, I can focus on the effect of *Alice* on growth and innovation outcomes for incumbent firms in the sample period from 2010 to 2019. I restrict to active firms with positive employment that received at least one patent between 2000 and 2009.[15] In section 3.4, I split the sample along the within-industry distribution of firm employment and patent portfolio size in 2010. Thus, I need to have a sufficient number of firms within each industry to define a patent portfolio size distribution and I restrict to firms in industries with at least 20 unique firms that received patent assignments between 2000 and 2009. This also increases the focus of the study on industries for which patenting is relevant and reduces the effect of outliers in the sample.

In the following, I generally refer to the estimated difference-in-difference coefficients as treatment effects. The average of the treatment variable in the full sample is 0.44 for treated firms and for the largest patent portfolio/employment size group in subsections 3.4 and 3.5 the average is 0.087 for treated firms. Multiplying the DiD coefficients with these averages gives average treatment effects for the treated.[16]

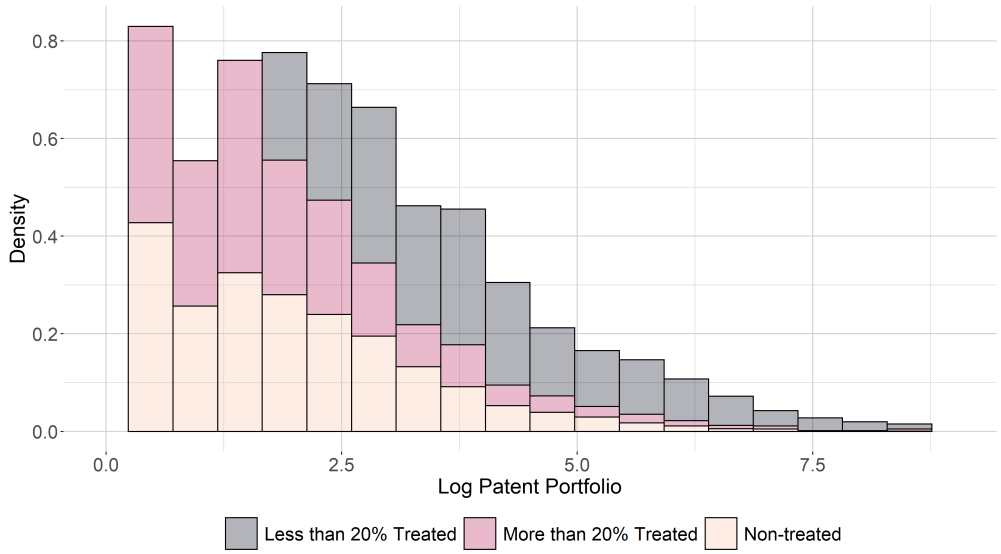
3.1. Balancing analysis

Before running the main analysis, it is important to understand how the *Alice* treatment is distributed across firms and industries and test whether covariates might have a confounding effect for identification.

[15] Note that restricting to firms with at least one establishment with positive employment does not mean that firms cannot drop out of the sample over the years due to, e.g., mergers or firm death. For robustness, I run the same analysis as below for continuing firms that have positive employment throughout the sample period. The results remain unchanged.

[16] See Callaway and Sant'Anna (2021) for more details on the assumptions and interpretation of average treatment effects in DiD settings with continuous treatment.

Figure 2 Patent portfolio distribution - 2010.



Note. Density-scale histogram of the cross-section of log patent portfolio size in 2010, grouped by *Alice*-treated patent shares. Patent portfolios are the unweighted total of all patents issued to the same U.S. company between 2000 and 2009. The three groups are portfolios with less than 20% of issued patents classified as *Alice*-treated by the NLP algorithm, portfolios with more than 20% of the patent portfolio treated, and portfolios without treated patents. The data are sourced from PatentsView.

First, I use PatentsView data to plot the histogram of the cross-sectional patent portfolio size distribution in 2010. Patent portfolios are the total count of patents assigned to the same company between 2000 and 2009. In figure 2, I distinguish between patent portfolios without treated patents, portfolios with a treatment share of less than 20%, and portfolios with more than 20% of patents being treated.

Larger shares of treated patents are related to smaller patent portfolios. With fewer patents the treatment shares are more discrete and can take large values. For example, a patent portfolio of two patents can have a treatment share of 0%, 50%, or 100%.[17] For larger patent portfolios, the distribution is more continuous. Patent portfolios with small treated patent shares are overall much larger than the average non-treated portfolio. This seems reasonable since firms with many patents are more likely to hold also a treated patent. Furthermore, as hinted in figure 1 and described in, e.g., Webb et al. (2018) and Forman and Goldfarb (2020), software patenting had been rising dramatically before *Alice* and the concentration of these patents has increased. The parallel trend assumption for my DiD method is not violated, though, if the distribution of the treatment variables is only correlated with the level of covariates. For the main analysis, I can include patent portfolio size as a covariate

[17] The coefficient in the pooled regression in table 1 is negative for the patent portfolio size, consistent with larger firms having overall smaller treatment shares.

Table 1 Cross-sectional regression - continuous treatment on covariates.

Log Patent PF	-0.002***								-0.006***
	(0.001)								(0.001)
Log Emp. - Lag		0.003***							0.006***
		(0.001)							(0.001)
Firm Age			0.000**						-0.001***
			(0.000)						(0.000)
Emp. Growth				0.004					0.009
				(0.003)					(0.016)
Entry Rate					0.000				-0.009
					(0.014)				(0.018)
Exit Rate						0.003			-0.005
						(0.010)			(0.013)
Job Creation Rate							0.003		-0.004
							(0.005)		(0.017)
Job Destruction Rate								-0.004	0.003
								(0.004)	(0.016)
Fixed Effects	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.
S.E. Cluster	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.
N	25,500	25,500	25,500	25,500	25,500	25,500	25,500	25,500	25,500
adj. R sq.	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13

Note. Cross-sectional regression of the treatment variables on firm covariates in 2010. The dependent variable is the share of *Alice*-treated patents among all patents issued to the firm between 2000 and 2009. 'Log PatentPF' is the log count of patents assigned to the firm between 2000 and 2009, 'Log Emp - Lag' is the log of total firm employment in 2009, all other covariates are 2010 values; definitions are in appendix A. Clustered standard errors are reported in parentheses, $^{*}p<0.1$,$^{**}p<0.05$,$^{***}p<0.01$.

in my TWFE regression model in equation 1 and match treated and control firms based on patent portfolio size.

I follow Goldsmith-Pinkham et al. (2020) and run in table 1 cross-sectional regressions of the treatment variable in 2010 on different covariates related to size and changes in firm outcomes.

Patent portfolio and firm size, measured in log terms, are correlated with treatment, as is firm age. With the unconditional average of the treatment variable around 0.05 in 2010, the magnitudes of the coefficients are overall small: for example, doubling the firm size increases the expected treatment value by just 0.002 to 0.004, between 5 and 10% of the unconditional average. The low R-squared values are also in line with the wide size distribution of treated patent portfolios in figure 2. Importantly, of the covariates related to changes in outcomes—employment growth, job creation and destruction rates, establishment entry and exit rates—none are significantly correlated with treatment.

Table 2 Correlation table covariates - 2010 cross-section.

Treatment	1								
Log PatentPF	-0.043	1							
Log Emp. - Lag	0.017	0.456	1						
Firm Age	-0.081	0.197	0.520	1					
Emp. Growth	0.007	-0.019	-0.088	-0.047	1				
Entry Rate	0.019	0.110	0.193	0.081	0.026	1			
Exit Rate	0.015	0.168	0.273	0.143	-0.115	0.199	1		
JC Rate	0.018	-0.013	-0.084	-0.118	0.631	0.265	0.085	1	
JD Rate	0.004	0.019	0.052	-0.027	-0.809	0.074	0.200	-0.151	1

Note. Correlation table for the cross-sectional variables used in table 1.

Overall, while treated firms are larger than non-treated firms, the share of treated patents is not strongly correlated with any covariate and seems to be uncorrelated with covariates related to changes in outcomes. Especially the latter is important for the parallel trend assumption. The correlation table for the covariates in table 2 confirms this: no correlation coefficient of any covariate with the treatment variable exceeds 0.081, much smaller than the correlation between, e.g., the patent portfolio and firm size of 0.46.

Finally, it is important to understand the distribution of treated firms across industries. In table 3, I show the five industries with the highest share of the treatment variable in 2010. The industry share of treatment is the sum of the treatment variables within each industry relative to the cross-sectional sum of the treated variable (this is related to the Rotemberg weights from Goldsmith-Pinkham et al. 2020). As further measures for the distribution of treatment, I include the average of the treatment variables within each industry, the industry share of treated firms, i.e., the number of treated firms in the industry relative to all treated firms across industries, and the industry share of all treated patent assignments.

The most treated industries all belong to service sectors related to software innovations. This is consistent with *Alice* invalidating software and business method patents. Computer Systems Design and Related Services (5415) stands out and accounts for 22% of the total treatment and 26% of all treated patents assigned to firms. Treated firms, i.e., firms with at least one treated patent assigned to them between 2000 and 2009, are less concentrated than the treatment variable itself; industry 5415 only accounts for 15.9% of all treated firms. The second most affected industry, Software Publishers (5112), accounts for ca. 10.4% of the total treatment, 11.7% of all treated patents, and 8.5% of treated firms. The treatment shares for all other industries are in the single digits. Thus, firms in the two most treated industries—5415 and 5112—account for approximately one-third of the overall treatment.

Table 3 Cross-industry distribution of treatment.

NAICS	Firms	Avg. Treatment	Share Treatment	Share Treated Firms	Share Patents	Share Treated Pat.
Computer Systems Design and Related Services (5415)	1,200	0.162	0.220	0.159	0.070	0.262
Software Publishers (5112)	450	0.210	0.104	0.085	0.022	0.117
Data Processing, Hosting, and Related Services (5182)	250	0.243	0.062	0.043	0.007	0.011
Management, Scientific, and Technical Consulting Services (5416)	450	0.091	0.047	0.028	0.002	0.006
Other Financial Investment Activities (5239)	100	0.288	0.034	0.019	0.001	0.003

Note. Cross-sectional distribution of treatment for the most treated industries. All variables are measured in 2010. Column two shows the number of patenting firms in the industry and column three the average of the treatment variable in the industry. Column four shows the sum of the treatment variable in the industry relative to the overall sum of the treatment across industries. Column five shows the number of treated firms in the industry relative to the total number of treated firms across industries. Treated firms have at least one treated patent assigned to them between 2000 and 2009. Column six shows the sum of patents received by firms in the industry relative to all patents assigned to firms between 2000 and 2009, and column seven the sum of treated patents assigned to firms in the industry relative to all treated patents assigned between 2000 and 2009.

Having treatment concentrated in software-related service sectors does not invalidate the parallel trend assumption, but I need to be careful and perform multiple robustness analysis. First, excluding the most treated industry 5415 does not change the results. Including four-digit-industry-year fixed effects as controls generally absorbs most of the treatment effects due to the relatively high concentration of the treatment variable in just two four-digit industries. However, including higher-level sector/two-digit-industry-year fixed effects does not change the results. Similarly, restricting the sample to service sectors does not qualitatively change the results but reduces the statistical significance of the DiD coefficients. Overall, treatment is correlated with industry but the main results, especially in section 3.5, are robust controlling for industry-wide trends.[18]

3.2. Patenting and innovation direction

I start the firm-level analysis by estimating the treatment effects of *Alice* on patenting measures related to innovation effort and direction for the full firm sample. In tables 4, 5, 7, and 6, I report the DiD results for measures of patenting quality, novelty, scope, and follow-on innovations. As a first-stage result, table 4 confirms that *Alice* decreases the ability of firms to patent inventions in software-related technologies: the annual number of issued patents has a negative treatment effect of 9%, with the same estimate for treated patents being 13.3%. The difference in treatment effects for forward-citation weighted patent issuances is even bigger (15.3% for all patent issuances and 35.5% for treated patents).

Table 5 adds to this result by showing that low-citation patents are most affected by *Alice*. Patent issuances in the top 10% of the forward citations distribution do not change. However, patents receiving few or no forward citations have a negative treatment effect of ca. 5%. Normalized forward citations show a negative treatment effect of 4.8%, around half the size of the treatment effect for overall issuances in table 4, meaning that the patenting quality has decreased less than the absolute number of issued patents. Software and business method patents are generally considered to be of low value (Hall and MacGarvie 2010), thus we would expect that the *Alice* shock affects lower-value patents by raising the patentability threshold of innovations. This is also consistent with the industry-level R&D results from appendix B.1 showing that overall innovation efforts do not change but rather the direction of innovation.

[18] It should be noted that my treatment is not fit to distinguish unconditional treatment intensity within industries. Besides the measurement errors from the NLP model and the patent-firm link, treatment takes more discrete and extreme values for smaller firms.

Table 4 Firm-level patenting - absolute measures.

	Patent Issue	Pat. Issue - Treated	Fwd. Cites	Fwd. Cites - Treated
Cont. Treatment * Post	-0.090***	-0.133***	-0.153***	-0.355***
	(0.021)	(0.016)	(0.041)	(0.034)
Fixed Effects	Firm+Year	Firm+Year	Firm+Year	Firm+Year
S.E. Cluster	Firm	Firm	Firm	Firm
N	197,000	197,000	197,000	197,000
adj. R sq.	0.85	0.83	0.73	0.67

Note. Difference-in-differences for absolute patenting outcomes. Observations are on the firm-year level for the year of patent issuance. The dependent variables are the total number of issued patents to the firm, the total number of *Alice*-treated patents issued to the firm, total forward citations to issued patents, and the total forward citations to issued *Alice*-treated patents. Dependent variables are in log terms and include '+1'. Treatment is the share of *Alice*-treated patents relative to all assigned patents to the firm between 2000 and 2009, the post-dummy is positive for years after 2014. Clustered standard errors are reported in parentheses, $^{*}p<0.1,^{**}p<0.05,^{***}p<0.01$.

Table 5 Firm-level patenting - quality measures.

	Top 1% Cited	Top 2-10% Cited	Cited Patents	Uncited Patents	Norm. Cites
Cont. Treatment * Post	-0.002	-0.012	-0.050***	-0.047***	-0.048**
	(0.004)	(0.010)	(0.017)	(0.016)	(0.021)
Fixed Effects	Firm+Year	Firm+Year	Firm+Year	Firm+Year	Firm+Year
S.E. Cluster	Firm	Firm	Firm	Firm	Firm
N	197,000	197,000	197,000	197,000	197,000
adj. R sq.	0.67	0.80	0.81	0.79	0.79

Note. Difference-in-differences for quality-adjusted patenting outcomes. Observations are on the firm-year level for the year of patent issuance. The dependent variables are the total number of issued patents in the top 1% of the forward citations distribution for patents in the same CPC group and filing year, the patents which are in the top 10% of the CPC-group-filing year forward citations distribution but not in the top 1%, the total number of patents that are outside of the top 10% of the CPC-group-filing year forward citations distribution but receive at least one citation, the number of patents that do not receive forward citations, and the total sum of forward citations normalized with the average number of forward citations in the CPC-group-filing year. Dependent variables are in log terms and include '+1'. Treatment is the share of *Alice*-treated patents relative to all assigned patents to the firm between 2000 and 2009, the post-dummy is positive for years after 2014. Clustered standard errors are reported in parentheses, $^{*}p<0.1,^{**}p<0.05,^{***}p<0.01$.

Tables 6 and 7 show that patented innovations become narrower in scope and more focused on incremental own innovation. The number of total claims decreases sharply with a treatment effect of -29.1%, while the normalized number of words in the first claim has a treatment effect of -6.2%, which is smaller than the overall decrease in issued patents. With relatively fewer claims and relatively more words in the first claim, the scope of patented inventions decreases (Kuhn and Thompson 2019).

Table 6 also shows a strong negative treatment effect of 24.6% for backward citations but no significant effect for self-citations. Backward citations proxy for the prior art and overall knowledge that inventions refer to, and self-citations indicate references to internal knowledge. This means that less external knowledge is referenced while prior innovations by

Table 6 Firm-level patenting - scope and innovation direction measures.

	Claims	Bwd. Cites	Self-Cites	New CPC Grp.	Wd. Claim	Norm. Wd. Claim
Cont. Treatment * Post	-0.291***	-0.246***	0.001	-0.014*	-14.840***	-0.062***
	(0.054)	(0.052)	(0.025)	(0.008)	(3.682)	(0.016)
Fixed Effects	Firm+Year	Firm+Year	Firm+Year	Firm+Year	Firm+Year	Firm+Year
S.E. Cluster	Firm	Firm	Firm	Firm	Firm	Firm
N	197,000	197,000	197,000	197,000	197,000	197,000
adj. R sq.	0.73	0.72	0.80	0.63	0.44	0.44

Note. Difference-in-differences for patenting scope and innovation direction outcomes. Observations are on the firm-year level for the year of patent issuance. The dependent variables are the total number of claims, the number of backward citations, the number of self-citations, the number of patents in new CPC groups the firm has not filed in before, the average number of words in the first claim, and the sum of the number of words in the first claim normalized with the average for patents in the same CPC-group-filing year. Dependent variables are in log terms and include '+1'. Treatment is the share of *Alice*-treated patents relative to all assigned patents to the firm between 2000 and 2009, the post-dummy is positive for years after 2014. Clustered standard errors are reported in parentheses, $^{*}p<0.1,^{**}p<0.05,^{***}p<0.01$.

the firm itself remain relevant. Following Akcigit and Kerr (2018), we can interpret this as a decrease in creative destruction since treated firms substitute new product development in favor of the improvement of existing products. This is further confirmed in table 7 by the negative treatment effect of 10.6% for follow-on citations and no significant effect for self-follow-on citations. Patents of treated firms receive fewer citations from outside inventors within the first three years after issuance but the number of citations by the firm itself does not significantly decrease, thus the patented inventions are less relevant for outside inventors compared to follow-on innovation by the firm itself.

Furthermore, *Alice* did not lead to the exploration of novel technologies since the number of patents filed in CPC groups the firm has not filed in before slightly decreases, as does the number of new CPC subgroup combinations appearing for the first time in the data and the number of product patents. A decrease in novelty and exploration is also consistent with the strong negative treatment effect of 12.5% for citations to scientific papers and basic research(Marx and Fuegi 2019).[19]

Overall, treated firms receive fewer low-citation patents after *Alice*. References to external knowledge and exploration decrease while references to current firm technologies become

[19] I find also strong negative treatment effects of -0.062 and -0.085, respectively, for technology proximity and the number of patents issued in CPC groups known to the firm (Known CPC Group). While the magnitude is close to the overall treatment effect for total patent issuance in table 4, these two results are partially mechanical since *Alice* is concentrated among specific CPC groups. With inventions in these CPC groups becoming patent ineligible following *Alice*, treated firms have to patent in different CPC groups than they used to. Neither effect is larger than the overall treatment effect for total patent issuances, further supporting that *Alice* did not induce exploration of new technologies.

Table 7 Firm-level patenting - novelty, process patents, and follow-on innovation measures.

	Bwd. Cites - Science	Follow-on Cites	Self-Follow-on Cites	Non-process Patents	Process Patents	New Combinations
Cont. Treatment * Post	-0.125***	-0.106***	0.000	-0.022**	-0.036***	-0.043*
	(0.035)	(0.025)	(0.016)	(0.011)	(0.010)	(0.024)
Fixed Effects	Firm+Year	Firm+Year	Firm+Year	Firm+Year	Firm+Year	Firm+Year
S.E. Cluster	Firm	Firm	Firm	Firm	Firm	Firm
N	197,000	197,000	197,000	197,000	197,000	197,000
adj. R sq.	0.79	0.79	0.79	0.82	0.80	0.70

Note. Difference-in-differences for patenting novelty, process patents, and follow-on innovation outcomes. Observations are on the firm-year level for the year of patent issuance. The dependent variables are the total number of backward citations to scientific articles, the sum of follow-on (forward) citations received within three years after issuance, the sum of follow-on (forward) citations by other patents issued to the same firm within three years, the number of non-process patents, the number process patents, and the number of pairwise CPC subgroup combinations that appear for the first time in the data. Dependent variables are in log terms and include '+1'. Treatment is the share of *Alice*-treated patents relative to all assigned patents to the firm between 2000 and 2009, the post-dummy is positive for years after 2014. Clustered standard errors are reported in parentheses, $^{*}p<0.1,^{**}p<0.05,^{***}p<0.01$.

more relevant, consistent with the marginal innovation direction shifting from creative destruction toward own innovation.

3.3. Firm-level growth

Tables 8 and 9 show the DiD regression results on the full firm sample for a broad range of LBD variables related to growth, employment, and establishment creation and destruction. Establishment entry and job creation decrease for treated firms after *Alice*. Job creation has a negative treatment effect of 12.5%, driven in part by the decrease in employment from entering establishments with a negative treatment effect of 7.8%. Interestingly, the wage share in table 8 has a negative treatment effect of 5.5%. As in Peters (2020) and Peters and Zilibotti (2021), we can interpret the decrease in the wage share as an increase in markups.[20] Taken together, the decrease in establishment entry and job creation and increase in markups are overall consistent with incumbent firms shifting away from the exploration of new technologies and toward improving existing products to increase the pricing power.[21] The idea behind this is rooted in Schumpeterian growth concepts and creative destruction: if a firm successfully explores a new technology, it becomes the monopolist in that product market and has to set up new establishments and create more jobs to serve the demand. If the firm instead improves the quality of its own products, it becomes more difficult for its competitors to close the quality gap, thus the firm can charge higher markups.

My results thus show a direct link between low-citation patents and establishment entry and job creation through creative destruction. Recent research by Abrams et al. (2013) and Abrams et al. (2019) provides evidence for an inverted U-shape for the relationship between forward citations and patent value with some high-value patents receiving few citations. The authors argue that low-citation patents add additional protection against infringement of other valuable inventions. I give empirical evidence for this idea by showing how low-citation *Alice* patents strengthen the patent protection of new ideas and incentivize companies to explore new products.

[20] I have revenue and wage bill data for around 80% of the LBD observations. The wage share is the ratio of the total wage bill to revenue. Following De Loecker and Warzynski (2012) and De Loecker et al. (2020), markups are the product of the output elasticity of labor and the inverse of the wage share, assuming labor is the only input. If the output elasticity is constant, I can measure changes in the log markups through our DiD setting using the log of the inverse of the wage share as outcome. In our TWFE model, the time and firm fixed effects control for the constant output elasticity.

[21] The results are robust for the same specification variations as described in section 3.5.

Table 8 Firm revenue, wage share, and market share.

	Real Rev.	Wage Share	Market Share
Cont. Treatment * Post	0.044	-0.055***	-0.039
	(0.028)	(0.020)	(0.032)
Fixed Effects	Firm+Year	Firm+Year	Firm+Year
S.E. Cluster	Firm	Firm	Firm
N	146,000	146,000	197,000
adj. R sq.	0.98	0.81	0.96

Note. Difference-in-differences for revenue, wage share, and market share. Observations are on the firm-year level. The dependent variables are the log of real revenue, the log of the wage share, and the log of the employment market share within the four-digit industry. Treatment is the share of *Alice*-treated patents relative to all assigned patents to the firm between 2000 and 2009, the post-dummy is positive for years after 2014. Clustered standard errors are reported in parentheses, $^{*}p<0.1$,$^{**}p<0.05$,$^{***}p<0.01$.

3.4. Split by firm size

To better understand which types of firms drive the effects in section 3.3, I split my sample along two dimensions: the size of the patent portfolio and total firm employment at the beginning of 2010, i.e., at the beginning of the pre-period. Within each four-digit NAICS industry, I split the sample into firms below the 50th percentile, firms in the 50th to 90th percentile, and firms in the top 10 percent of the patent and employment size distribution, respectively. Table 10 shows the means of relevant variables within the different brackets. As described in section 3.1, firms are very heterogeneous in terms of patent portfolio and employment size; while the smallest firms have on average less than 40 employees and between one and four patents in their portfolio, companies in the top brackets have several thousands of employees and hundreds of patents in their portfolios. Particularly firms that are above the 90th percentile of the patent portfolio and also above the 90th percentile of employment size distribution stand out, accounting for almost half of all new establishment entries and close to 80% of all patent issuances. Patenting seems less relevant for small firms in *Alice* affected industries.[22]

I can apply the same regression models as above within each patent portfolio/employment size bracket. This improves sample balancing since I now explicitly control for firm employment and patent portfolio size.[23] In tables 11 and 12, I report the treatment effects for patent

[22] This might also explain why firm creation and destruction on the industry level do not change in appendix B: small firms with one establishment make up a large share of entering and exiting establishments but hardly file patents in the first place.

[23] No coefficient is significant if I perform the same cross-sectional regression of treatment on covariates as in table 1 for the largest patent portfolio/employment size group.

Table 9 Firm establishments.

	Estabs (Pos. Emp.)	Emp.	Entry	Entry Emp.	Exit	Exit Emp.	Job Creation	Job Destruction
Cont. Treatment * Post	0.007	0.000	-0.032	-0.078**	-0.012	-0.023	-0.125***	0.075*
	(0.014)	(0.028)	(0.020)	(0.038)	(0.019)	(0.038)	(0.047)	(0.043)
Fixed Effects	Firm+Year	Firm+Year	Firm+Year	Firm+Year	Firm+Year	Firm+Year	Firm+Year	Firm+Year
S.E. Cluster	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm
N	197,000	197,000	197,000	197,000	197,000	197,000	197,000	197,000
adj. R sq.	0.98	0.97	0.64	0.53	0.69	0.59	0.66	0.65

Note. Difference-in-differences for establishment-related outcomes. Observations are on the firm-year level. The dependent variables are the total number of establishments with positive employment, total employment, the number of entering establishments, the employment of entering establishments, the number of exiting establishments, the employment in t-1 of exiting establishments, and total job creation and destruction. Dependent variables are in log terms and include '+1', except for the log of the count of establishments and total employment which are required to be positive. Treatment is the share of *Alice*-treated patents relative to all assigned patents to the firm between 2000 and 2009, the post-dummy is positive for years after 2014. Clustered standard errors are reported in parentheses, $^{*}p < 0.1$,$^{**}p < 0.05$,$^{***}p < 0.01$.

Table 10 Variable means by patent portfolio and employment size.

Patent PF Tier	Emp. Tier	N	Treatment	Estabs.	Emp.	Patent PF	Mkt. Share	Entry	Entry Emp.	JC	Wage Share	Pat. Issue	Follow-on	Self-F/O	Bwd. Cites	Self-Cites
<0.5 Pctl.	<0.5 Pctl.	37,000	0.018	1.63	45.96	1.55	0.000	0.04	0.84	4.77	0.29	0.22	0.32	0.11	7.57	0.27
<0.5 Pctl.	0.5-0.9 Pctl.	19,000	0.020	8.74	369.80	2.85	0.001	0.59	10.02	37.79	0.28	0.60	0.92	0.37	15.62	0.88
<0.5 Pctl.	>0.9 Pctl.	2,100	0.025	85.66	2,194.00	3.40	0.009	6.50	65.74	246.90	0.26	1.02	1.73	0.92	30.00	1.45
0.5-0.9 Pctl.	<0.5 Pctl.	53,500	0.036	1.46	35.45	4.36	0.000	0.04	0.70	3.76	0.30	0.59	0.69	0.25	13.86	0.88
0.5-0.9 Pctl.	0.5-0.9 Pctl.	53,000	0.041	10.42	518.80	10.86	0.001	0.62	12.58	55.08	0.30	1.69	2.34	0.86	46.88	3.07
0.5-0.9 Pctl.	>0.9 Pctl.	9,900	0.052	136.70	5,735.00	28.95	0.012	10.34	202.50	690.80	0.27	6.61	7.99	3.37	140.30	9.89
>0.9 Pctl.	<0.5 Pctl.	3,200	0.016	1.62	39.96	48.53	0.000	0.06	0.50	3.70	0.30	5.33	4.73	1.50	79.96	6.43
>0.9 Pctl.	0.5-0.9 Pctl.	8,400	0.025	28.40	909.10	124.20	0.003	2.58	27.24	99.33	0.26	15.27	20.44	10.10	292.00	48.00
>0.9 Pctl.	>0.9 Pctl.	10,500	0.032	229.60	10,950.00	857.00	0.028	19.31	337.90	1,235.00	0.23	115.40	158.60	63.51	2,295.00	259.40

Note. Firm-year variable means by patent portfolio and firm employment size groups. Firms are grouped based on their rank in the patent portfolio and firm employment size distribution within their four-digit NAICS industry. Within each industry, firms are ranked below the 50th percentile, between the 50th and 90th percentile, or above the 90th percentile based on the total number of patents assigned between 2000 and 2009, and independent from this based on total firm employment in 2009. N is the total number of observations, treatment is the share of *Alice*-treated patents relative to all assigned patents to the firm between 2000 and 2009, all other variables are defined in appendix A.

Table 11 Firm-level patent issuances by patent portfolio and employment size group.

Patent Issue	<0.5 pctl. emp.	0.5-0.9 pctl. emp.	>0.9 pctl. emp.
<0.5 pctl. pat.	-0.030	-0.015	-0.131
	(0.054)	(0.093)	(0.106)
0.5-0.9 pctl. pat.	-0.049***	-0.108***	0.001
	(0.018)	(0.033)	(0.100)
>0.9 pctl. pat.	-0.522	-0.869**	-1.256***
	(0.430)	(0.343)	(0.296)
Fixed Effects	Firm+Year	Firm+Year	Firm+Year
S.E. Cluster	Firm	Firm	Firm

Note. Difference-in-differences for total annual patent issuances for different patent portfolio and firm employment size groups. Within each four-digit NAICS industry, firms are ranked below the 50th percentile, between the 50th and 90th percentile, or above the 90th percentile based on the total number of patents assigned between 2000 and 2009 (row), and independent from this based on total firm employment in 2009 (column). For each of the possible nine sub-sample groups, the difference-in-differences estimates for the same setting as in table 4 are reported. Dependent variables are in log terms and include '+1'. Clustered standard errors are reported in parentheses, $^{*}p<0.1,^{**}p<0.05,^{***}p<0.01$.

issuances and establishment entry for different patent portfolio and employment size groups. The treatment effects for patenting are negative and significant for multiple patent portfolio/employment size groups since *Alice* limited the overall patentability of software-related innovations. However, the largest patent portfolio/employment size group—firms above the 90th percentile of the patent portfolio as well as employment size distribution—are most affected with a negative treatment effect of 1.256. Only this group has a significant treatment effect of -1.069 for establishment entry. Thus, limiting the patentability of innovations is most relevant for the largest patenting firms and leads to a decrease in establishment entry for these firms, consistent with reduced exploration of new technologies that would require new business locations.[24]

To test the parallel trend assumption for the most treated group of firms above the 90th percentile of the patent portfolio as well as employment size distribution, I use model 2 to

[24] For robustness, I implement an alternative sample split with absolute patent portfolio and firm employment size cutoffs. I group firms in patent portfolio brackets counting the number of patents assigned to them between 2000 and 2009: 1 to 5 patents, 6 to 19, or more than 20 patents. Independent from this, I define employment size brackets based on the total firm employment in 2009: 1 to 99 employees, 100 to 999 employees, or more than 1000 employees. The cutoffs are selected to allow for a sufficient sample size within each bracket and to set the cutoffs for the top brackets close to the 90-95th percentile in the patent portfolio and employment size distribution. The results confirm that the decrease in patenting is more widely distributed across the different size brackets. The treatment effect for establishment entry is also here only significant for the largest patent portfolio and employment size bracket with a magnitude close to effect for the largest patent portfolio/employment size group in table 12.

Table 12 Firm-level establishment entry by patent portfolio and employment size group.

Entry	<0.5 pctl. emp.	0.5-0.9 pctl. emp.	>0.9 pctl. emp.
<0.5 pctl. pat.	0.095	0.079	-0.145
	(0.065)	(0.066)	(0.173)
0.5-0.9 pctl. pat.	0.013	-0.029	-0.222
	(0.012)	(0.026)	(0.135)
>0.9 pctl. pat.	0.004	0.053	-1.069***
	(0.043)	(0.231)	(0.335)
Fixed Effects	Firm+Year	Firm+Year	Firm+Year
S.E. Cluster	Firm	Firm	Firm

Note. Difference-in-differences for annual establishment entry for different patent portfolio and firm employment size groups. Within each four-digit NAICS industry, firms are ranked below the 50th percentile, between the 50th and 90th percentile, or above the 90th percentile based on the total number of patents assigned between 2000 and 2009 (row), and independent from this based on total firm employment in 2009 (column). For each of the possible nine sub-sample groups, the difference-in-differences estimates for the same setting as in table 9 are reported. Dependent variables are in log terms and include '+1'. Clustered standard errors are reported in parentheses, $^{*}p<0.1,^{**}p<0.05,^{***}p<0.01$.

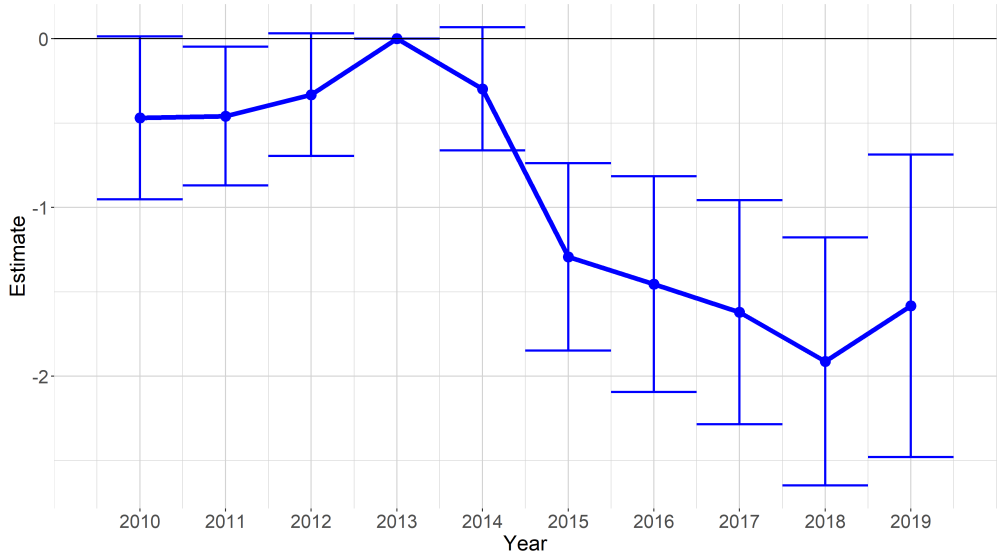
plot in figures 3 and 4 the event study results for patent issuances and establishment entry. I find no pre-trend or inverted "Ashenfelter's Dip" for either outcome.[25] We also see that the treatment effect for patent issuances sets in immediately in 2015. The effect on establishment entry gradually increases starting in 2016. This is consistent with the reduced patentability of innovations causing establishment entry to decrease over time. Note also that LBD data measure employment in March and the *Alice* decision was in June 2014, thus, the first full year after *Alice* is 2016.

3.5. Large patenting firms

The results in section 3.4 show that firms above the 90th percentile of the patent portfolio and employment size distribution within industries are the most affected by *Alice*. In this section, I implement different variations of the DiD model for this group of firms to estimate robust treatment effects. Tables 13, 14, and 15 report treatment effects for four model variants: the base specification without covariates, the base specification with standard errors clustered on industry level, and two specifications that include log firm employment size, log patenting portfolio size, and firm age as covariates. The first specification with covariates interacts the

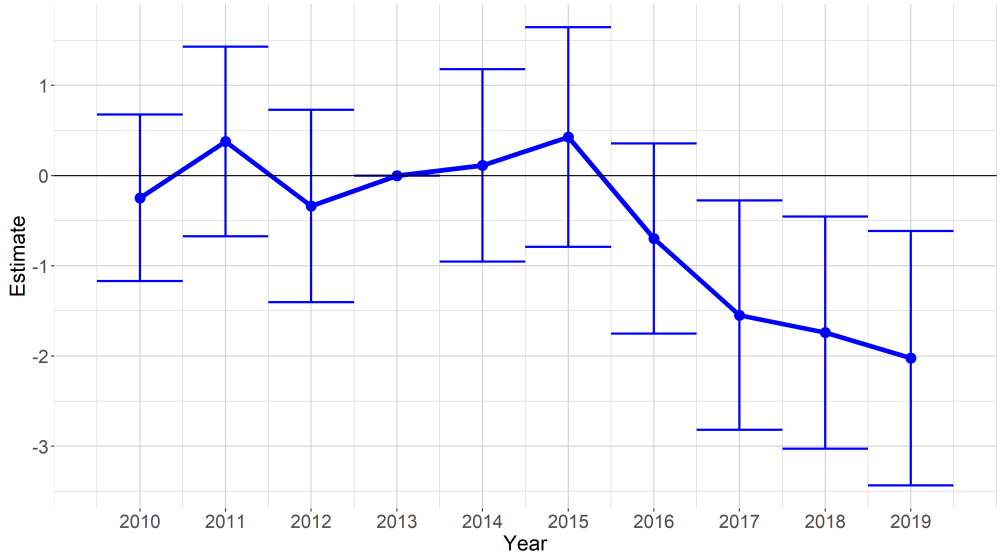
[25] The coefficient for 2011 in figure 3 is significantly negative. However, no other coefficient before 2015 is significant, no trend is visible, and the magnitude is very small compared to the post-treatment effect.

Figure 3 Event-study - Patent Issue - >90th pctl. pat. - >90th pctl. emp.



Note. Event study for total annual patent issuances for the group of firms above the 90th percentile of the within-industry patent portfolio size and firm employment size distribution in 2009 as defined in table 10. Observations are on the firm-year level. Treatment effects are estimated with the event-study model in equation 2. Dependent variables are in log terms and include '+1'. Coefficients for the interaction of the treatment variable with the year dummies are shown with error bars for the 95% confidence interval. Standard errors are clustered on the firm level.

Figure 4 Event-study - Entry - >90th pctl. pat. - >90th pctl. emp.



Note. Event study for establishment entry for the group of firms above the 90th percentile of the within-industry patent portfolio size and firm employment size distribution in 2009 as defined in table 10. Observations are on the firm-year level. Treatment effects are estimated with the event-study model in equation 2. Dependent variables are in log terms and include '+1'. Coefficients for the interaction of the treatment variable with the year dummies are shown with error bars for the 95% confidence interval. Standard errors are clustered on the firm level.

controls with the post-dummy, the second with year dummies. These are the only significant covariates in the cross-sectional regression in table 3, including them as controls might help

to improve the precision of the DiD coefficient estimate (Angrist and Pischke 2009).[26] The average of the treatment variable for the treated firms is 0.087, thus the average treatment effect for the treated can be calculated by multiplying the difference-in-difference coefficients with this average.[27]

In appendix C, I show event-study results for the base specification and confirm that there are no pre-trends, except for the follow-on citations outcome with a marginally significant coefficient for 2010. This might be related to fewer patents being issued after *Alice* that could cite a related patent. Nevertheless, the treatment effects for follow-on citations are highly significant after 2015 and the magnitude remains much larger than for the self-follow-on citations.[28]

The results in table 13 show a large decrease in patent issuances following *Alice* with treatment effects between -1.40 and -1.26. We also see a clear gap in the magnitudes of the treatment effects for backward citations (-2.56 to -2.47) and self-citation (-0.97 to -0.75) as well as for follow-on citations (-1.53 to -1.42) and self-follow-on citations (-0.83 to -0.64). Overall, treated firms reference their own patents relatively more after *Alice*, consistent with focusing more on internal knowledge relative to external knowledge. Following Akcigit and Kerr (2018), I can interpret this as treated firms shifting away from creative destruction innovation toward own innovation.

Table 14 shows negative treatment effects for establishment entry (-1.07 to -0.64), employment from establishment entry (-1.84 to -1.29), and job creation (-1.57 to -1.14).[29] *Alice* has a negative effect on the exploration of new technologies. The limited patentability of inventions leads treated firms to introduce fewer new technologies and thus diminishes the creation of new establishments and jobs.

There is a negative treatment effect for job destruction, which is significant at the 10%-level for the specification with standard errors clustered on the industry level. In table 9,

[26] Since the treatment is continuous and the shock nationwide, there are few group-level controls that are useful to include. The results are robust to excluding observations in industry 5415, winsorizing the top 1% of observations, and restricting to firms with non-missing observations for the entire sample period of 2010 to 2019. The results remain largely unchanged when including three-digit-industry-year fixed effects. Furthermore, I can use Coarsened Exact Matching (CEM) on log firm employment, log patenting portfolio size, firm age, and sector dummies to balance the covariate distribution of treated and control firms at the beginning of the pre-period in 2010. The results remain largely unchanged when using the CEM-balanced sample for the analysis.

[27] As before, I generally refer to the estimated difference-in-difference coefficients as treatment effects.

[28] Event study results for the full sample also show no pre-trend except for follow-on citations.

[29] Establishment entry-related results are robust to controlling for re-timed entry and excluding re-activated establishments that temporarily had no employees for one to seven years before entry.

Table 13 Firm-level patenting - >90th pctl. pat. & >90th pctl. emp. - specification variation.

Patent Issue				
Cont. Treatment * Post	-1.256***	-1.256***	-1.399***	-1.401***
	(0.296)	(0.266)	(0.295)	(0.295)
N	10,500	10,500	10,500	10,500
adj. R sq.	0.90	0.90	0.90	0.90
Backward Citations				
Cont. Treatment * Post	-2.470***	-2.470***	-2.557***	-2.564***
	(0.466)	(0.369)	(0.471)	(0.471)
N	10,500	10,500	10,500	10,500
adj. R sq.	0.81	0.81	0.81	0.81
Self-Citations				
Cont. Treatment * Post	-0.747*	-0.747**	-0.966**	-0.970**
	(0.423)	(0.381)	(0.422)	(0.422)
N	10,500	10,500	10,500	10,500
adj. R sq.	0.86	0.86	0.86	0.86
Follow-on				
Cont. Treatment * Post	-1.417***	-1.417***	-1.532***	-1.534***
	(0.323)	(0.270)	(0.317)	(0.317)
N	10,500	10,500	10,500	10,500
adj. R sq.	0.86	0.86	0.86	0.86
Self-Follow-on				
Cont. Treatment * Post	-0.638**	-0.638***	-0.823***	-0.826***
	(0.308)	(0.228)	(0.303)	(0.303)
N	10,500	10,500	10,500	10,500
adj. R sq.	0.84	0.84	0.85	0.85
Fixed Effects	Firm+Year	Firm+Year	Firm+Year	Firm+Year
S.E. Cluster	Firm	Industry	Firm	Firm
Controls	N	N	Post-interaction	Year-interaction

Note. Difference-in-differences for various model specifications - patenting outcomes. The dependent variables are defined as in tables 4 and 7. Results are shown for the group of firms above the 90th percentile of the within-industry patent portfolio size and firm employment size distribution in 2009 as defined in table 10. The first column shows results for the base DiD setting. Column two clusters standard errors on the industry level without additional covariates. Columns three and four include as covariates the log firm employment size, log patent portfolio size, and firm age, all measured in 2009. Column three interacts the covariates with the post-dummy, column four interacts the covariates with year dummies. Clustered standard errors are reported in parentheses, $^{*}p < 0.1, ^{**}p < 0.05, ^{***}p < 0.01$.

however, there is a positive treatment effect for job destruction in the full sample. Some firms in the mid-tiers of the patent portfolio and employment size distribution have positive treatment effects for job destruction (which are not robust, though, to the specification variations in this section). Thus, *Alice* had a different effect for smaller treated firms: the limited enforceability of some of their patents leads to more competition in some of their products, increasing job destruction. The firms in the largest patent portfolio/employment

size group, however, have on average 857 patents assigned to them between 2000 and 2009; even with a shock to the enforceability of software-related patents, they still hold large patent portfolios of valid patents that are unlikely to be more successfully challenged after *Alice* (see also Choi and Gerlach 2015). Instead, the insignificant negative treatment effect for job destruction for this group of firms is more consistent with the transition into a new equilibrium with less creative destruction. As shown in figures 3 and 4 in section 3.4, the decrease in patent issuances after *Alice* leads to a gradual reduction in establishment entry. If firms move over time into a new equilibrium with less creative destruction, we would expect also job destruction to eventually follow. Since the treatment effect for job destruction in the largest patent portfolio/employment size group is only marginally significant, the new equilibrium might not have been reached yet by the end of the sample period in 2019.[30]

Finally, table 15 shows that the negative treatment effects for establishment count and total employment only survive for the specifications without controls. The high serial correlation of establishment count and firm employment might explain the insignificant estimates when including size-related covariates. The magnitudes of the treatment effects without covariates are in line with the treatment effect for the market share (-0.64 to -0.57). Thus, the decrease in creative destruction following *Alice* eventually leads to smaller market shares for treated firms.[31] Since I find no clear positive treatment effect for the market share in any other patent portfolio/employment size group, it seems likely that the treated firms in the largest patent portfolio/employment size group lose market share to a wider range of competitors.[32]

[30] I do not report treatment effects for establishment exit since I cannot control for re-timed and re-activated establishment exits as for establishment entries. The coverage of the relevant re-timing flag for exiting establishments is much lower than for entering establishments. Re-timing is overall more of a challenge for exits, as can be seen in spikes for establishment exits around Economic Census years. Since a large part of the treatment effect is driven by the last two sample years (after the last Economic Census in 2017), I focus on job destruction which is more directly observable for those years. Nevertheless, I find a significant negative treatment trend for establishment exits (-0.206 per year after 2014) and employment of exiting establishments (-0.381 per year after 2014) for the largest patent portfolio/employment size group of firms, consistent with a long-term decrease in creative destruction.

[31] Note that overall the effects are too small to change the industry-wide HHI and other concentration measures.

[32] The treatment effect for the market share is larger than what we would expect from a decrease in new job creation alone (the job creation rate is close to 10% of total firm employment and the magnitude of the treatment effect for job creation is only around twice as large as the treatment effect for the market share). One explanation could be that treated firms sell operations related to new technology exploration through M&A activities. This is consistent with unreported results that show an increase in the number of establishments that change ownership after *Alice* (see also Garcia-Macia et al. 2019). Overall, the relatively large decrease in market share for treated firms is consistent with the interpretation that *Alice* reduces new technology exploration.

Table 14 Firm-level growth - >90th pctl. pat. & >90th pctl. emp. - specification variation.

Entry				
Cont. Treatment * Post	-1.069***	-1.069***	-0.652**	-0.640**
	(0.335)	(0.323)	(0.315)	(0.314)
N	10,500	10,500	10,500	10,500
adj. R sq.	0.59	0.59	0.60	0.60
Entry Emp.				
Cont. Treatment * Post	-1.840***	-1.840***	-1.289**	-1.267**
	(0.544)	(0.496)	(0.535)	(0.535)
N	10,500	10,500	10,500	10,500
adj. R sq.	0.46	0.46	0.46	0.46
Job Creation				
Cont. Treatment * Post	-1.571***	-1.571***	-1.161**	-1.146*
	(0.594)	(0.510)	(0.591)	(0.591)
N	10,500	10,500	10,500	10,500
adj. R sq.	0.53	0.53	0.53	0.54
Job Destruction				
Cont. Treatment * Post	-0.710	-0.710*	-0.342	-0.333
	(0.505)	(0.383)	(0.508)	(0.509)
N	10,500	10,500	10,500	10,500
adj. R sq.	0.56	0.56	0.56	0.56
Fixed Effects	Firm+Year	Firm+Year	Firm+Year	Firm+Year
S.E. Cluster	Firm	Industry	Firm	Firm
Controls	N	N	Post-interaction	Year-interaction

Note. Difference-in-differences for various model specifications - growth outcomes. The dependent variables are defined as in table 9. Results are shown for the group of firms above the 90th percentile of the within-industry patent portfolio size and firm employment size distribution in 2009 as defined in table 10. The first column shows results for the base DiD setting. Column two clusters standard errors on the industry level without additional covariates. Columns three and four include as covariates the log firm employment size, log patent portfolio size, and firm age, all measured in 2009. Column three interacts the covariates with the post-dummy, column four interacts the covariates with year dummies. Clustered standard errors are reported in parentheses, $^{*}p < 0.1, ^{**}p < 0.05, ^{***}p < 0.01$.

Table 15 also shows a positive treatment effect of 0.22 to 0.25 for markups. The estimate is insignificant for one of the specifications with covariates and only significant at the 5%-level for standard errors clustered at the industry level. It should be noted, though, that the sample size is more than 20% smaller for markups than for the other outcomes since the last available year is 2018, and that the magnitude of the estimates does not change much across different specifications. The increase in markups for treated firms is consistent with the shift of innovation efforts toward own innovation. Improving the already existing products increases the quality gap to the next-best competitor and increases the pricing power of incumbent firms, thus allowing for higher markups.

Table 15 Firm-level size and markup - >90th pctl. pat. & >90th pctl. emp. - specification variation.

Estabs. (Pos. Emp.)				
Cont. Treatment * Post	-0.425*	-0.425***	-0.385	-0.383
	(0.241)	(0.147)	(0.244)	(0.244)
N	10,500	10,500	10,500	10,500
adj. R sq.	0.95	0.95	0.95	0.95
Employment				
Cont. Treatment * Post	-0.461**	-0.461**	-0.355	-0.353
	(0.225)	(0.211)	(0.220)	(0.220)
N	10,500	10,500	10,500	10,500
adj. R sq.	0.92	0.92	0.92	0.92
Markup				
Cont. Treatment * Post	0.246*	0.246**	0.219	0.221*
	(0.128)	(0.111)	(0.134)	(0.134)
N	7,900	7,900	7,900	7,900
adj. R sq.	0.85	0.85	0.85	0.85
Market Share				
Cont. Treatment * Post	-0.637**	-0.637**	-0.570**	-0.567**
	(0.270)	(0.290)	(0.271)	(0.271)
N	10,500	10,500	10,500	10,500
adj. R sq.	0.88	0.88	0.88	0.88
Fixed Effects	Firm+Year	Firm+Year	Firm+Year	Firm+Year
S.E. Cluster	Firm	Industry	Firm	Firm
Controls	N	N	Post-interaction	Year-interaction

Note. Difference-in-differences for various model specifications - size outcomes. The dependent variables are defined as in tables 8 and 9; Markup is the inverse of Wage Share (in log-terms). Results are shown for the group of firms above the 90th percentile of the within-industry patent portfolio size and firm employment size distribution in 2009 as defined in table 10. The first column shows results for the base DiD setting. Column two clusters standard errors on the industry level without additional covariates. Columns three and four include as covariates the log firm employment size, log patent portfolio size, and firm age, all measured in 2009. Column three interacts the covariates with the post-dummy, column four interacts the covariates with year dummies. Clustered standard errors are reported in parentheses, $^{*}p<0.1$,$^{**}p<0.05$,$^{***}p<0.01$.

Overall, all results are robust and support the main finding: after *Alice* limited the patent protection of inventions, large incumbent firms substitute technology exploration and creative destruction for own innovation, leading to a decrease in job creation, more references to internal knowledge, and an increase in markups.

4. Conclusion

Patents, market entry, and innovation are closely related: patents provide a temporary monopoly on inventions to incentivize the exploration of new ideas. New technologies such as computer software challenge the balance between what defines a patent-eligible innovation and what is an abstract idea giving unfair competitive advantages if patented. The Supreme

Court decision in *Alice Corp. v. CLS Bank International* limited the patent eligibility of software and business method claims and significantly changed the patenting landscape for inventors. I use this shock to identify the effects of patentability on firm innovation and growth.

I develop a novel NLP-based approach to classify patents if they are at risk of being invalid under *Alice*. I identify treated firms and industries in restricted-use U.S. Census data and use a difference-in-differences approach to estimate the causal effect of reduced patentability on growth and innovation outcomes. I find no significant change in the entry or exit of new ventures, but a significant decrease of 3 to 4% in new establishment creation by incumbents. On the firm level, I find negative treatment effects for overall patent issuances and job creation but positive effects for markups. Patent citations to internal knowledge become more important. Large patenting firms drive the effects and focus on improving the products they already have in place rather than introducing new technologies after the court decision limited the patentability of new inventions. Overall, my results show that patents incentivize incumbents to explore new technologies, thereby contributing to overall growth through creative destruction.

My study opens a new perspective on how innovation, patentability, and growth are empirically connected. I contribute to multiple strains of literature with my novel identification and microdata-based evidence for Schumpeterian growth with heterogeneous innovations. There are several limitations to this analysis, though: patent applications typically take two to three years to be granted and my sample only has five years of post-treatment observations. Thus, my analysis does not fully capture the long-run effect of *Alice*. *Alice* is also industry specific to software-oriented service sectors, and less than 3% of all patents are treated. Nevertheless, *Alice* is relevant for affected firms and significantly changes their innovation and growth trajectory. This raises new questions about how patentability might affect investment, financing, and labor demand. Which type of workers benefits from the shift toward own innovation? Are fixed assets substitutes or complements for patent-protected inventions? Does creative destruction raise funding costs? The purpose of this study is to set the framework for future studies to answer these questions and deepen our understanding of patentability, innovation, and growth.

References

Abadie, A., S. Athey, G. W. Imbens, and J. Wooldridge (2017, November). When Should You Adjust Standard Errors for Clustering? Working Paper 24003, National Bureau of Economic Research. Series: Working Paper Series.

Abrams, D. S., U. Akcigit, and J. Grennan (2013, November). Patent Value and Citations: Creative Destruction or Strategic Disruption? Working Paper 19647, National Bureau of Economic Research. Series: Working Paper Series.

Abrams, D. S., U. Akcigit, G. Oz, and J. G. Pearce (2019, March). The Patent Troll: Benign Middleman or Stick-Up Artist? Working Paper 25713, National Bureau of Economic Research. Series: Working Paper Series.

Aghion, P., U. Akcigit, and P. Howitt (2014, January). Chapter 1 - What Do We Learn From Schumpeterian Growth Theory? In P. Aghion and S. N. Durlauf (Eds.), *Handbook of Economic Growth*, Volume 2 of *Handbook of Economic Growth*, pp. 515–563. Elsevier.

Aghion, P., A. Bergeaud, T. Boppart, P. J. Klenow, and H. Li (2019, November). A Theory of Falling Growth and Rising Rents. Working Paper 26448, National Bureau of Economic Research.

Aghion, P., P. Howitt, and S. Prantl (2015, September). Patent rights, product market reforms, and innovation. *Journal of Economic Growth 20*(3), 223–262.

Akcigit, U. and S. T. Ates (2019, April). What Happened to U.S. Business Dynamism? Working Paper 25756, National Bureau of Economic Research. Series: Working Paper Series.

Akcigit, U. and W. R. Kerr (2018, March). Growth through Heterogeneous Innovations. *Journal of Political Economy 126*(4), 1374–1443. Publisher: The University of Chicago Press.

Angrist, J. and J.-S. Pischke (2009, January). *Mostly Harmless Econometrics* (1 edition). Princeton University Press.

Appel, I., J. Farre-Mensa, and E. Simintzi (2019, September). Patent trolls and startup employment. *Journal of Financial Economics 133*(3), 708–725.

Arrow, K. (1962, January). Economic Welfare and the Allocation of Resources for Invention. Working Paper c2144, National Bureau of Economic Research.

Arts, S. and L. Fleming (2018, December). Paradise of Novelty—Or Loss of Human Capital? Exploring New Fields and Inventive Output. *Organization Science 29*(6), 1074–1092.

Balsmeier, B., L. Fleming, and G. Manso (2017, March). Independent boards and innovation. *Journal of Financial Economics 123*(3), 536–557.

Bena, J., H. Ortiz-Molina, and E. Simintzi (2021, October). Shielding Firm Value: Employment Protection and Process Innovation. *Journal of Financial Economics*.

Bertrand, M., E. Duflo, and S. Mullainathan (2004, February). How Much Should We Trust Differences-In-Differences Estimates?*. *The Quarterly Journal of Economics 119*(1), 249–275.

Bessen, J. and R. M. Hunt (2007). An Empirical Look at Software Patents. *Journal of Economics & Management Strategy 16*(1), 157–189. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1530-9134.2007.00136.x.

Boldrin, M. and D. K. Levine (2013, February). The Case Against Patents. *Journal of Economic Perspectives 27*(1), 3–22.

Callaway, B. and P. H. C. Sant'Anna (2021, December). Difference-in-Differences with multiple time periods. *Journal of Econometrics 225*(2), 200–230.

Chien, C. V. (2016). Software Patents As A Currency, Not Tax, On Innovation. *Berkeley Technology Law Journal 31*(3), 1669–1723. Publisher: [University of California, Berkeley, University of California, Berkeley, School of Law].

Chien, C. V. and J. Y. Wu (2018, October). Decoding Patentable Subject Matter. SSRN Scholarly Paper ID 3267742, Social Science Research Network, Rochester, NY.

Choi, J. P. and H. Gerlach (2015). Patent pools, litigation, and innovation. *The RAND Journal of Economics 46*(3), 499–523.

Chow, M. C., T. C. Fort, C. Goetz, N. Goldschlag, J. Lawrence, E. R. Perlman, M. Stinson, and T. K. White (2021, May). Redesigning the Longitudinal Business Database. Working Paper w28839, National Bureau of Economic Research.

Cockburn, I. M. and M. J. MacGarvie (2009). Patents, Thickets and the Financing of Early-Stage Firms: Evidence from the Software Industry. *Journal of Economics & Management Strategy 18*(3), 729–773. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1530-9134.2009.00228.x.

Cockburn, I. M. and M. J. MacGarvie (2011, April). Entry and Patenting in the Software Industry. *Management Science 57*(5), 915–933. Publisher: INFORMS.

Cohen, L., U. G. Gurun, and S. D. Kominers (2019, September). Patent Trolls: Evidence from Targeted Firms. *Management Science*, mnsc.2018.3147.

Contigiani, A. (2020, April). Experimentation and Appropriability in Early-Stage Ventures: Evidence from the Software Industry. SSRN Scholarly Paper ID 3282261, Social Science Research Network, Rochester, NY.

Craig, J. A. (2017). Deconstructing Wonderland: Making Sense of Software Patents in a Post-Alice World Patent Law. *Berkeley Technology Law Journal 32*(Annual Review Issue), 359–378.

Daily, J. E. (2017). Alice's Aftermath: Changes in Patentee Behavior since Alice v. CLS Bank Bridging the Gap between the Federal Courts and the United States Patent & Trademark Office: The Journal of Science and Technology Law Symposium. *Boston University Journal of Science and Technology Law 23*(2), 284–303.

De Loecker, J., J. Eeckhout, and G. Unger (2020, May). The Rise of Market Power and the Macroeconomic Implications. *The Quarterly Journal of Economics 135*(2), 561–644.

De Loecker, J. and F. Warzynski (2012, October). Markups and Firm-Level Export Status. *American Economic Review 102*(6), 2437–2471.

DiNizo, A. M. J. (2018). From Alice to Bob: The Patent Eligibility of Blockchain in a post CLS Bank World. *Case Western Reserve Journal of Law, Technology and the Internet 9*, 1–28.

Dreisigmeyer, D., N. Goldschlag, M. Krylova, W. Ouyang, and E. Perlman (2018, July). Building a Better Bridge: Improving Patent Assignee-Firm Links. CES Technical Notes Series, Center for Economic Studies, U.S. Census Bureau.

Dugan, B. (2018). Mechanizing Alice: Automating the Subject Matter Eligibility Test of Alice v. CLS Bank. *University of Illinois Journal of Law, Technology & Policy 2018*(1), 33–80.

Forman, C. and A. Goldfarb (2020, June). Concentration and Agglomeration of IT Innovation and Entrepreneurship: Evidence from Patenting. Working Paper w27338, National Bureau of Economic Research.

Fort, T. and S. Klimek (2018, June). The Effects of Industry Classification Changes on US Employment Composition. Working Paper, U.S. Census Bureau, Center for Economic Studies.

Frésard, L., G. Hoberg, and G. M. Phillips (2020, July). Innovation Activities and Integration through Vertical Acquisitions. *The Review of Financial Studies 33*(7), 2937–2976.

Galasso, A. and M. Schankerman (2015, February). Patents and Cumulative Innovation: Causal Evidence from the Courts *. *The Quarterly Journal of Economics 130*(1), 317–369.

Gallini, N. T. (1992). Patent Policy and Costly Imitation. *The RAND Journal of Economics 23*(1), 52–63. Publisher: [RAND Corporation, Wiley].

Gans, J. S. and S. Stern (2003, February). The product market and the market for "ideas": commercialization strategies for technology entrepreneurs. *Research Policy 32*(2), 333–350.

Garcia-Macia, D., C.-T. Hsieh, and P. J. Klenow (2019). How Destructive Is Innovation? *Econometrica 87*(5), 1507–1541. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.3982/ECTA14930.

Gilbert, R. and C. Shapiro (1990). Optimal Patent Length and Breadth. *The RAND Journal of Economics 21*(1), 106–112. Publisher: [RAND Corporation, Wiley].

Goldschlag, N., T. J. Lybbert, and N. J. Zolas (2016, March). An 'Algorithmic Links with Probabilities' Crosswalk for USPC and CPC Patent Classifications with an Application Towards Industrial Technology Composition. SSRN Scholarly Paper ID 2749287, Social Science Research Network, Rochester, NY.

Goldsmith-Pinkham, P., I. Sorkin, and H. Swift (2020, August). Bartik Instruments: What, When, Why, and How. *American Economic Review 110*(8), 2586–2624.

Graham, S. J. H., C. Grim, T. Islam, A. C. Marco, and J. Miranda (2018). Business Dynamics of Innovating Firms: Linking U.S. Patents with Administrative Data on Workers and Firms. SSRN Scholarly Paper ID 3224865, Social Science Research Network, Rochester, NY.

Graham, S. J. H., A. C. Marco, and R. Miller (2015, November). The USPTO Patent Examination Research Dataset: A Window on the Process of Patent Examination. SSRN Scholarly Paper ID 2702637, Social Science Research Network, Rochester, NY.

Gutiérrez, G. and T. Philippon (2016, December). Investment-less Growth: An Empirical Investigation. Working Paper w22897, National Bureau of Economic Research.

Hall, B. H., G. v. Graevenitz, and C. Helmers (2021, April). Technology entry in the presence of patent thickets. *Oxford Economic Papers 73*(2), 903–926.

Hall, B. H., A. Jaffe, and M. Trajtenberg (2005). Market Value and Patent Citations. *The RAND Journal of Economics 36*(1), 16–38. Publisher: [RAND Corporation, Wiley].

Hall, B. H. and M. MacGarvie (2010, September). The private value of software patents. *Research Policy 39*(7), 994–1009.

Hall, R. E. (2018, May). New Evidence on the Markup of Prices over Marginal Costs and the Role of Mega-Firms in the US Economy. Working Paper 24574, National Bureau of Economic Research.

Haltiwanger, J., R. Jarmin, R. Kulick, J. Miranda, V. Penciakova, and C. Tello-Trillo (2019, August). Firm-level Revenue Dataset. Working Paper 19-02, Center for Economic Studies, U.S. Census Bureau. Publication Title: CES Technical Notes Series.

Hansen, C. B. (2007, October). Generalized least squares inference in panel and multilevel models with serial correlation and fixed effects. *Journal of Econometrics 140*(2), 670–694.

Harhoff, D., F. Narin, F. M. Scherer, and K. Vopel (1999, August). Citation Frequency and the Value of Patented Inventions. *The Review of Economics and Statistics 81*(3), 511–515.

Hoberg, G. and G. Phillips (2016, October). Text-Based Network Industries and Endogenous Product Differentiation. *Journal of Political Economy 124*(4), 1423–1465.

Jarmin, R. S. and J. Miranda (2002). The Longitudinal Business Database. *SSRN Electronic Journal*.

Ke, Z. T., B. T. Kelly, and D. Xiu (2019, July). Predicting Returns with Text Data. SSRN Scholarly Paper ID 3389884, Social Science Research Network, Rochester, NY.

Kelly, B., D. Papanikolaou, A. Seru, and M. Taddy (2021, September). Measuring Technological Innovation over the Long Run. *American Economic Review: Insights 3*(3), 303–320.

Kerr, W. R. and R. Nanda (2009). Democratizing entry: Banking deregulations, financing constraints, and entrepreneurship$. *Journal of Financial Economics 94*(1), 124–149.

Kesan, J. P. and R. Wang (2020, March). Eligible Subject Matter at the Patent Office: An Empirical Study of the Influence of Alice on Patent Examiners and Patent Applicants. SSRN Scholarly Paper ID 3556216, Social Science Research Network, Rochester, NY.

Klemperer, P. (1990). How Broad Should the Scope of Patent Protection Be? *The RAND Journal of Economics 21*(1), 113–130. Publisher: [RAND Corporation, Wiley].

Klenow, P. J. and H. Li (2021, May). Innovative Growth Accounting. *NBER Macroeconomics Annual 35*, 245–295. Publisher: The University of Chicago Press.

Kogan, L., D. Papanikolaou, A. Seru, and N. Stoffman (2017, May). Technological Innovation, Resource Allocation, and Growth*. *The Quarterly Journal of Economics 132*(2), 665–712.

Kong, N., U. Dulleck, A. B. Jaffe, S. Sun, and S. Vajjala (2020, September). Linguistic Metrics for Patent Disclosure: Evidence from University Versus Corporate Patents. Working Paper 27803, National Bureau of Economic Research. Series: Working Paper Series.

Kuhn, J., K. Younge, and A. Marco (2020). Patent citations reexamined. *The RAND Journal of Economics 51*(1), 109–132. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/1756-2171.12307.

Kuhn, J. M. and N. C. Thompson (2019, January). How to Measure and Draw Causal Inferences with Patent Scope. *International Journal of the Economics of Business 26*(1), 5–38.

Lee, J.-S., N. Kim, and Z.-T. Bae (2019, December). The effects of patent litigation involving NPEs on firms' patent strategies. *Technological Forecasting and Social Change 149*, 119758.

Lemley, M. A. and R. Feldman (2016, May). Patent Licensing, Technology Transfer, and Innovation. *American Economic Review 106*(5), 188–192.

Lemley, M. A. and C. Shapiro (2005, June). Probabilistic Patents. *Journal of Economic Perspectives 19*(2), 75–98.

Lerner, J. and A. Seru (2021, July). The Use and Misuse of Patent Data: Issues for Finance and Beyond. *The Review of Financial Studies* (hhab084).

Lerner, J., A. Seru, N. Short, and Y. Sun (2021, July). Financial Innovation in the 21st Century: Evidence from U.S. Patents. Working Paper 28980, National Bureau of Economic Research. Series: Working Paper Series.

Loughran, T. and B. Mcdonald (2011). When Is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks. *The Journal of Finance 66*(1), 35–65.

Lu, Q., A. Myers, and S. Beliveau (2017, November). USPTO Patent Prosecution Research Data: Unlocking Office Action Traits. SSRN Scholarly Paper ID 3024621, Social Science Research Network, Rochester, NY.

Lybbert, T. J. and N. J. Zolas (2012, September). Getting Patents and Economic Data to Speak to Each Other: An 'Algorithmic Links with Probabilities' Approach for Joint Analyses of Patenting and Economic Activity. SSRN Scholarly Paper ID 2161982, Social Science Research Network, Rochester, NY.

Marx, M. and A. Fuegi (2019, November). Reliance on Science: Worldwide Front-Page Patent Citations to Scientific Articles. SSRN Scholarly Paper ID 3331686, Social Science Research Network, Rochester, NY.

Miller, S. P., A. Aravind, B. Bengfort, C. De La Cerda, M. Dragoni, K. Gibson, A. Itai, C. Johnson, D. Kannappan, E. Kehoe, H. Kim, K. Mladinich, R. Pinho, J. Polansky, and B. Weissenberg (2018). Who's Suing Us? Decoding Patent Plaintiffs since 2000 with the Stanford NPE Litigation Dataset. *Stanford Technology Law Review 21*(2), 235–275.

Moser, P. (2011, April). Innovation Without Patents - Evidence from the World Fairs. SSRN Scholarly Paper ID 930241, Social Science Research Network, Rochester, NY.

Noel, M. and M. Schankerman (2013). Strategic Patenting and Software Innovation. *The Journal of Industrial Economics 61*(3), 481–520.

Nordhaus, W. D. (1969). An Economic Theory of Technological Change. *The American Economic Review 59*(2), 18–28. Publisher: American Economic Association.

Peters, M. (2020). Heterogeneous Markups, Growth, and Endogenous Misallocation. *Econometrica 88*(5), 2037–2073. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.3982/ECTA15565.

Peters, M. and F. Zilibotti (2021, October). Creative Destruction, Distance to Frontier, and Economic Development. Working Paper 29333, National Bureau of Economic Research. Series: Working Paper Series.

Ren, X. and R. Duprez (2019, May). Financial impact of legal practice on software patent subject matter eligibility: Effectiveness of a workable standard. *The Journal of High Technology Management Research 30*(1), 50–58.

Sampat, B. and H. L. Williams (2019, January). How Do Patents Affect Follow-On Innovation? Evidence from the Human Genome. *American Economic Review 109*(1), 203–236.

Schumpeter, J. A. (1942). Capitalism, Socialism, and Democracy. SSRN Scholarly Paper ID 1496200, Social Science Research Network, Rochester, NY.

Stroud, J. and D. M. Kim (2017). Debugging Software Patents after Alice. *South Carolina Law Review 69*(1), 177–220.

Toole, A. A. and N. A. Pairolero (2020, April). Adjusting to Alice. Working Paper 3, United States Patent and Trademark Office.

Tran, J. L. (2015). Software Patents: A One-Year Review of Alice v. CLS Bank. *Journal of the Patent and Trademark Office Society 97*(3), 532–550.

Tran, J. L. (2016). Two Years after Alice v. CLS Bank. *Journal of the Patent and Trademark Office Society 98*(3), 354–373.

Tran, J. L. and J. S. Benevento (2019). Alice at Five. SSRN Scholarly Paper ID 3454922, Social Science Research Network, Rochester, NY.

Wagner, S. and I. Cockburn (2010, March). Patents and the survival of Internet-related IPOs. *Research Policy 39*(2), 214–228.

Webb, M., N. Short, N. Bloom, and J. Lerner (2018, July). Some Facts of High-Tech Patenting. Working Paper w24793, National Bureau of Economic Research.

Online Appendix A: Variable definitions

In the following section, I describe measures for the novelty, scope, and quality of innovations using data from PatentsView.

I follow Balsmeier et al. (2017) and measure the number of backward citations, i.e., the number of citations to prior patents, and self-citations, citations to prior patents of the same firm. Backward citations proxy for the prior art and overall knowledge that inventions refer to, and self-citations indicate references to internal knowledge of the firm. [33]

The number of citations patents receive (forward citations) is a measure of innovation quality. I differentiate patents further into four groups based on the distribution of forward citations for patents within the same filing year and CPC group: I count the number of patents that are in the top 1% of the forward citations distribution, that are in the 2–10% percentile, that are below the top decile but have received at least one forward citation, and patents that have received no citations. According to Balsmeier et al. (2017), this allows us to measure breakthrough, important, incremental, and failed inventions. I further follow Lerner and Seru (2021) and normalize forward citations by the average number of citations received by patents filed in the same year and CPC group, which provides a more continuous measure of quality.[34]

I build on Galasso and Schankerman (2015) and measure follow-on citation as the total forward citations received by patents within three years after issuance. Since PatentsView data are available until 2021, citations within three years after issuance can be measured up to 2018 and partially for 2019. I follow the empirical approach in the working paper to Akcigit and Kerr (2018) and distinguish follow-on citations made by the firm itself as self-follow-on citations. The purpose of follow-on measures is to understand how important inventions are for the firm's own innovation efforts (own innovation) and other innovators (creative destruction).

[33] PatentsView creates time-consistent unique identifiers for assignees of granted patents based on a string-matching protocol. I use these assignee data to identify self-citations, technology proximity, and new and known CPC groups rather than the patent-firm link by Dreisigmeyer et al. (2018) since the coverage of PatentsView is much longer (1976 to 2021 compared to 2000 to 2019). The overlap between the firms identified by PatentsView and the patent-firm link in Census data is very high: 84% of assignees in PatentsView have a unique firm identifier in the patent-firm link, and 81% of firms in the patent-firm link have a unique assignee-ID in PatentsView. 94% of the patents covered by the patent-firm link have a unique assignee in PatentsView data. In total, at least 3 million patents are covered by both the patent-firm link and PatentsView.

[34] I use filing years to normalize the forward citations distribution to measure the quality of eligible innovations at the time of the invention rather than the time of grant.

I measure the novelty of inventions by counting the pairwise combinations of CPC subgroups that appear for the first time in the patent database in patents issued to the firm, following Arts and Fleming (2018). The total number of claims is related to the scope of patented inventions. Furthermore, I measure the scope of patents by counting the number of words in the first claim and normalize by the average number of words in the first claim for patents filed in the same CPC group and year, following Kuhn and Thompson (2019). The first claim is usually the broadest claim of a patent. As the scope of technology covered by a patent becomes more narrow and specific, more words are used to describe the invention. Thus, broader patents have fewer words in the first claim. I include data from Marx and Fuegi (2019) counting the number of citations to scientific papers (non-patent literature, NPL citations) as a measure of how focused on basic research inventions are. I also include data from Bena et al. (2021) to measure process and product innovations.

Tables 16 and 17 summarize the definitions of economic and innovation variables at the firm level. Abbreviations are in brackets following the variable name. I refer to the current year as 't'.

Variables names in tables may begin with 'Log' to clarify they are in log terms. If patent measures apply to the *Alice*-treated patents, the variable names in tables are followed by '-Treated'.

Table 16 Economic variable definitions.

Variable	Definition	Data source/Reference
Employment (Emp.)	Sum of total employment of all firm establishments in t.	LBD
Employment Growth (Emp. Growth)	Change in employment from t-1 to t over the average employment of the firm for t and t-1.	LBD
Entry	Sum of all firm establishments where t-1 employment is zero and current year employment is >0.	LBD, Chow et al. (2021)
Entry Employment (Entry Emp.)	Sum of employment in t of entering firm establishments.	LBD, Chow et al. (2021)
Entry Rate	Entry relative to the average number of establishments of the firm in t and t-1. The number of establishments in t-1 is calculated by subtracting from the establishment count in t the number of entering establishments and adding the number of exiting establishments.	LBD, Chow et al. (2021)
Establishments (Estabs.)	Sum of all establishments of the firm in t. "An establishment is a single physical location at which business is conducted or services or industrial operations are performed."[a]	LBD
Establishments (Pos. Emp.)	Sum of all firm establishments in t with at least one active employee.	LBD
Exit	Sum of all firm establishments where employment in t-1 is >0 and employment in t is 0.	LBD. Chow et al. (2021)
Exit Employment	Sum of employment in t-1 of exiting firm establishments.	LBD, Chow et al. (2021)
Exit Rate	Exit relative to the average number of establishments of the firm in t and t-1. The number of establishments in t-1 is calculated by subtracting from the establishment count in t the number of entering establishments and adding the number of exiting establishments.	LBD, Chow et al. (2021)
Firm Age	Difference between the observation year and the first year the firm is observed with positive employment.	LBD, Chow et al. (2021)
Firm Death	Dummy which is one in the last year that the firm is observed with positive employment.	LBD, Chow et al. (2021)
Firm Size	Firm total employment in t for in-scope establishments for BDS.	LBD, Chow et al. (2021)
Job Creation (JC)	Sum of positive employment change of firm establishments from t-1 to 1.	LBD, Chow et al. (2021)
Job Creation Rate (JC Rate)	Job creation relative to the average employment of the firm for t and t-1.	LBD, Chow et al. (2021)
Job Destruction (JD)	Sum of negative employment change of firm establishments from t-1 to t.	LBD, Chow et al. (2021)
Job Destruction Rate (JD Rate)	Job destruction relative to the average employment of the firm for t and t-1.	LBD, Chow et al. (2021)
Market Share (Mkt. Share)	Firm employment in t relative to the total employment in t for all firms in the same four-digit NAICS industry.	LBD
Markup	Inverse of the wage share.	LBD, Haltiwanger et al. (2019)
Real Revenue (Real Rev.)	Total revenue of the firm in t adjusted for inflation to the base year 2009.	LBD, Haltiwanger et al. (2019)
Real Revenue Growth	Change of real revenue from t-1 to t relative to the average real revenue of the firm for t-1 and t.	LBD, Haltiwanger et al. (2019)
t and t-1	Current year (t) and previous year (t-1).	
Wage Share (W/S)	Ratio of total payroll to nominal revenue of the firm in t.	LBD, Haltiwanger et al. (2019)

[a] See `www.census.gov/programs-surveys/susb/about/glossary.html`

Table 17 Innovation variable definitions.

Variable	Definition	Data source/Reference
Average Words First Claim (Wd. Claim)	Average number of words in the first claim in patents issued to the firm in t.	PatentsView, Kuhn and Thompson (2019)
Backward Citations (Bwd. Cites)	Sum of citations by patents issued to the firm in t.	PatentsView
Backward Citations to Scientific Articles (Bwd. Cites - Science)	Sum of citations to scientific articles by patents issued to the firm in t.	Marx and Fuegi (2019)
Cited Patents	Sum of all patents issued to the firm in t that are below the 90th percentile of the forward citations distribution within the same filing year and CPC group but received at least one forward citation.	PatentsView, Balsmeier et al. (2017)
Claims	Sum of all independent claims in patents issued to the firm in t.	PatentsView, Balsmeier et al. (2017)
Follow-on Citations (Follow-on Cites; Follow-on)	Sum of all forward citations to patents issued to the firm in t within three years after issuance.	PatentsView
Forward Citations (Fwd. Cites)	Sum of all citations to patents issued to the firm in t.	PatentsView, Balsmeier et al. (2017)
Known CPC Group	Sum of patents issued to the firm in t in CPC groups the firm has filed in before.	PatentsView, Balsmeier et al. (2017)
New CPC Group (New CPC Grp.)	Sum of patents issued to the firm in t in CPC groups the firm has not filed in before.	PatentsView, Balsmeier et al. (2017)
New CPC Subgroup Combinations (New Combinations)	Sum of total pairwise CPC subgroup combinations by patents issued to the firm in t that have not appeared before in the data.	PatentsView, Arts and Fleming (2018)
Non-process Patents	Sum of patents issued to the firm in t that have no process claims.	Bena et al. (2021)
Normalized Words First Claim (Norm. Wd. Claim)	Sum of the number of words in the first claim in patents issued to the firm in t normalized by the respective average of words in the first claim for patents in the same filing year and CPC group.	PatentsView, Kuhn and Thompson (2019)
Normalized Forward Citations (Norm. Cites)	Sum of forward citations to patents issued to the firm in t relative to the respective average number of forward citations to patents in the same filing year and CPC group.	PatentsView, Lerner and Seru (2021)
Patent Issue (Pat. Issue)	Sum of all patents issued to the firm in t.	PatentsView
Patent Portfolio (Patent PF)	Sum of all issued patents to the firm between 2000 and 2009.	PatentsView
Process Patents	Sum of patents issued to the firm in t that have all independent claims classified as a process claim.	Bena et al. (2021)
Self-Citations (Self-Cites)	Sum of citations by patents issued to the firm in t to other patents assigned to the same firm.	PatentsView, Balsmeier et al. (2017)
Self-Follow-on Citations (Self-Follow-on Cites; Self-F/O)	Sum of all forward citations to patents issued to the firm in t by other patents issued to the same firm within three years.	PatentsView
Technology Proximity	Average cosine similarity of patents issued to the firm in t to prior patents of the same firm. Cosine similarity is calculated for the count vector of CPC groups for a given filing year relative to the count vector of CPC groups of all patents filed by the same firm over the prior 10 years.	PatentsView, Balsmeier et al. (2017)
Top 1% Forward Citations (Top 1% Cited)	Sum of all patents issued to the firm in t that are above the 99th percentile of the forward citations distribution within the same filing year and CPC group.	PatentsView, Balsmeier et al. (2017)
Top 2% to 10% Forward Citations (Top 2-10% Cited)	Sum of all patents issued to the firm in t that are between the 90th and 99th percentile of the forward citations distribution within the same filing year and CPC group.	PatentsView, Balsmeier et al. (2017)
Treatment Share	Sum of *Alice*-treated patents issued to the firm between 2000 and 2009 relative to all patents issued to the same firm between 2000 and 2009.	PatentsView
Uncited Patents	Sum of all patents issued to the firm in t that have not received any forward citations.	PatentsView, Balsmeier et al. (2017)

Online Appendix B: Industry results

I can use the *Alice* shock to ask how patenting affects new firm creation and firm exit on the industry level. As described in section 2.2, I use the vintage-consistent four-digit NAICS codes from Fort and Klimek (2018) to define industries and exclude the same industries as Kerr and Nanda (2009). I aggregate annual establishment entry and exit from the LBD to measure new firm creation and destruction on the industry level.

I limit the sample to industries that have at least 20 distinct firms with patent assignments between 2000 and 2009. This allows testing the parallel trend assumption for a pre-period and avoids issues with selection into treatment before *Alice*. While the precise number of patenting firms in an industry is not relevant for the result, having too few patents and firms with patents in an industry poses issues for identification: as with the firm level, the treatment variable can take extreme values if the numbers of patents involved become too small and I might include industries in the treatment or control groups for which patenting is not important.[36]

The treatment variable is the share of treated patents relative to all patents assigned to firms in the four-digit NAICS industry between 2000 and 2009. While this treatment definition is based on the patent portfolios of incumbents, it identifies industries for which *Alice* is relevant and the patentability of inventions is important.[37]

I use the difference-in-differences regression model from equation 1 with observations on the industry-year level between 2010 and 2019. I follow Kerr and Nanda (2009) and use as outcomes the log of the sum of entering and exiting establishments and total employment of entering and exiting establishments. As in Kerr and Nanda (2009), I fill zero values on the

[36] For example, if we can find only one firm with one single treated patent assigned to it the industry-level treatment would be 100%. Despite the high quality of the patent-firm match in Dreisigmeyer et al. (2018) with a precision of more than 90%, measurement error might distort results if there are too few distinct firms used to identify treated industries.

[37] For robustness, I implement an alternative approach that is independent of firm-level links and aggregating firm patent portfolios to industry level by using the Algorithmic Links with Probabilities (ALP) patent crosswalk from Goldschlag et al. (2016) and Lybbert and Zolas (2012). This patent crosswalk is based on NLP methods and provides a direct link between CPC/USPC classes and industry NAICS codes by matching keywords from patent abstracts to industry and product classification descriptions. This means patent classes can be matched according to how important the described technologies are for industries. First, I count the total number of issued patents and treated patents between 2000 and 2009 for each CPC group. I then use the NAICS-CPC bridge from the ALP patent crosswalk to calculate the probability-weighted sum of issued patents and the sum of treated patents for each four-digit NAICS industry. I restrict to industries above the median number of total issued patents to make sure that patents are relevant for the industries in the sample. I then define as treated industries those with an above-median share of treated patents relative to all issued patents. The main results remain unchanged with no increase in single-unit entry and a strong decrease in multi-unit establishment entry.

Table 18 Industry entry.

	Entry	Entry Emp.	MU Entry	SU Entry	MU Entry Emp.	SU Entry Emp.
Cont. Treatment * Post	-0.300	-0.627*	-1.135***	0.245	-0.924**	0.054
	(0.409)	(0.371)	(0.391)	(0.501)	(0.443)	(0.444)
Fixed Effects	Ind. + Year	Ind. + Year	Ind. + Year	Ind. + Year	Ind. + Year	Ind. + Year
S.E. Cluster	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.
N	1,800	1,800	1,800	1,800	1,800	1,800
adj. R sq.	0.99	0.95	0.96	0.99	0.89	0.96

Note. Difference-in-differences for industry entry. Observations are on the industry-year level. The dependent variables are the total sum of all entering establishments, the sum of overall employment of entering establishments, the sum of all multi-unit establishment entries, the sum of single-unit entries, the sum of employment of multi-unit establishment entries, and the sum of employment of single-unit entrants. Single-unit (SU) entries are new firm formations with a single establishment, multi-unit (MU) entries are new facilities created by existing firms with other continuing locations. All dependent variables are in log terms. All regressions include a dummy for zero-values of the dependent variable; the respective observations are replaced with one. Treatment is the share of *Alice*-treated patents relative to all assigned patents between 2000 and 2009 to firms in the industry, the post-dummy is positive for years after 2014. Clustered standard errors are reported in parentheses, $^{*}p<0.1$,$^{**}p<0.05$,$^{***}p<0.01$.

Table 19 Industry exit.

	Exit	Exit Emp.	MU Exit	SU Exit	MU Exit Emp.	SU Exit Emp.
Cont. Treatment * Post	-0.025	-0.136	-0.048	0.128	-0.049	0.044
	(0.399)	(0.480)	(0.405)	(0.415)	(0.610)	(0.373)
Fixed Effects	Ind. + Year	Ind. + Year	Ind. + Year	Ind. + Year	Ind. + Year	Ind. + Year
S.E. Cluster	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.
N	1,800	1,800	1,800	1,800	1,800	1,800
adj. R sq.	0.99	0.95	0.97	0.99	0.88	0.97

Note. Difference-in-differences for industry exit. Observations are on the industry-year level. The dependent variables are the total sum of all exiting establishments, the sum of overall previous year employment of exiting establishments, the sum of all multi-unit establishment exits, the sum of single-unit exits, the sum of previous year employment of multi-unit establishment exits, and the sum of previous year employment of single-unit exits. Single-unit (SU) exits are firms with one location ceasing operations, multi-unit (MU) exits are the closing of facilities by firms with other continuing locations. All regressions include a dummy for zero-values of the dependent variable; the respective observations are replaced with one. Treatment is the share of *Alice*-treated patents relative to all assigned patents between 2000 and 2009 to firms in the industry, the post-dummy is positive for years after 2014. Clustered standard errors are reported in parentheses, $^{*}p<0.1$,$^{**}p<0.05$,$^{***}p<0.01$.

left-hand side to one and add a dummy variable to control for those cases.[38] Furthermore, I can distinguish between single-unit and multi-unit firms. Single-unit firms have only one establishment, thus single-unit entries and exits are in most cases the same as new venture creation and firm destruction.[39] Multi-unit entry and exit refer to the entry and exit of establishments belonging to firms that have multiple other existing establishments and thus are predominantly large incumbents.

[38] The results remain unchanged if I use instead 'log(y + 1)' on the LHS as in section 3.

[39] Only a small share of entering and exiting firms have multiple establishments at birth or death and most firms in the economy are single-unit establishments.

Table 20 Patenting by entrants.

	Count Issue	Sum Issue	Count Issue - Treated	Sum Issue - Treated
Cont. Treatment * post-dummy	-1.763***	-0.971	-3.792***	-5.546***
	(0.650)	(1.078)	(0.924)	(1.421)
Fixed Effects	Ind. + Year	Ind. + Year	Ind. + Year	Ind. + Year
S.E. Cluster	Ind.	Ind.	Ind.	Ind.
N	1,800	1,800	1,800	1,800
adj. R sq.	0.87	0.78	0.81	0.84

Note. Difference-in-differences for the patenting of entrants. Observations are on the industry-year level. The dependent variables are the count of entering establishments with at least one patent assigned in the year of entry, the sum of patents issued to entrants, the count of entrants with at least one *Alice*-treated patents assigned in the year of entry, and the sum of *Alice*-treated patents assigned to entrants. All dependent variables are in log terms. All regressions include a dummy for zero-values of the dependent variable; the respective observations are replaced with one. Treatment is the share of *Alice*-treated patents relative to all assigned patents between 2000 and 2009 to firms in the industry, the post-dummy is positive for years after 2014. Clustered standard errors are reported in parentheses, $^{*}p<0.1,^{**}p<0.05,^{***}p<0.01$.

Tables 18 and 19 show the DiD results for industry-level entry and exit. Overall, establishment entry decreases, especially weighted by employment. The treatment effect of -0.63 log points is almost completely driven by a decrease in multi-unit entry with a treatment effect of -1.14 for multi-unit entry and -0.92 for employment-weighted multi-unit entry. With an average continuous treatment of 0.034 for treated industries, this translates to a decrease in multi-unit entry of 3.1 to 3.8%. In no specification does single-unit entry change following *Alice*. This result is robust to the inclusion of sector-year fixed effects and sector trends, using non-retimed entry, and using a binary treatment variable for industries above the median share of treated patents. Thus, *Alice* has no effect on small firm entry or exit, but reduces new establishment creation by large multi-unit firms.

In table 20, I follow the same econometric approach as above and aggregate patent assignments to entering establishments at the industry level. I measure the overall count of entering establishments that received at least one patent in the year of entry and the total number of patents received. I also count the entrants with *Alice*-treated patents assigned and the sum of treated patents assigned to entering establishments.[40] I find significant negative treatment effects for all measures except the total number of patents issued to entrants. The average treatment effect of 5.98%[41] for the count of entrants with at least one patent assigned to them is even larger than the effect for multi-unit entry in table 18, thus the decrease in

[40] The results change little if I use the patents received in the year before or after entry instead. The only coefficient that changes is the treatment effect for the total number of patents assigned to entrants which becomes significantly negative.

[41] -1.76 * 3.4% (DiD coefficient * average of the treatment variable for treated industries)

patenting entrants drives the decrease in industry entry. Overall, our second important result in this section shows that the decrease in establishment entry is accompanied by a decrease in entrants that receive patents in the year of entry due to the limited patentability of *Alice* related inventions.

Taking stock, we find that *Alice* led to a reduction in new establishment creation by large firms. New ventures seem to enter with new products no matter whether they own a patent or not. Thus, in section 3, I focus on how incumbent firms are changing their growth and innovation strategy following *Alice*.

B.1. R&D results

I use survey responses in the Business R&D and Innovation Survey (BRDIS) to analyze how innovation efforts and direction change after *Alice*. BRDIS is designed to be the 'primary source of information on research and development performed or funded by businesses within the United States.'[42] The results thus are supposed to reflect broader trends and are most useful for industry analysis.

I use BRDIS data for the years 2010 to 2019 and merge survey responses to the industry-level patenting statistics from appendix B to identify treated industries. I adapt the regression model 1 and weight observations by their respective survey weights.[43] The weighted observations are representative of industry-wide innovation efforts and thus treatment effects measure industry-level responses to *Alice*.[44] It should be noted, though, that the survey responses go through multiple stages of editing and adjustments before weights are assigned.[45] Also the survey design has changed multiple times over the years, leading to potential inconsistencies over time. The results, thus, should be taken as evidence for the directional effect of *Alice* on innovation outcomes rather than as precise estimates.

Tables 21 and 22 show results for worldwide and domestic R&D expenses relative to sales and in absolute Dollar terms. I find no significant treatment effects in any specification, thus *Alice* did not change innovation effort in treated industries.

[42] See `www.nsf.gov/statistics/srvyberd/#sd`

[43] Aggregating to industry-level first would overstate the precision of observations measured with few survey responses, nevertheless I find also here no significant change in R&D spending.

[44] Since large innovative firms are more likely to be sampled by BRDIS, I can link certain firms from the main panel in section 3 directly to their R&D responses. This is problematic, though, since BRDIS is not longitudinal and would add sampling error to any firm-level analysis. Furthermore, BRDIS samples heavily from manufacturing firms and the DiD setting only works reasonably well with multiple pre- and post-period observations per firm, restricting the number of observations even more. Nevertheless, the results from tables 21, 22, and 23 remain unchanged with no decrease in R&D efforts and a directional change away from research.

[45] The results remain unchanged when using unweighted observations.

Table 21 Worldwide R&D spending.

	Log R&D Exp.	Log R&D Perf.	Log R&D Emp.	R&D Exp. to Sales	R&D Perf. to Sales	R&D Emp. to Emp.
Cont. Treatment * Post	-0.237	-0.202	0.170	-0.001	0.003	-0.006
	(0.391)	(0.402)	(0.137)	(0.010)	(0.011)	(0.024)
Fixed Effects	Ind. + Year	Ind. + Year	Ind. + Year	Ind. + Year	Ind. + Year	Ind. + Year
S.E. Cluster	Ind.	Ind.	Ind.	Ind.	Ind.	Ind.
N	167,000	167,000	167,000	167,000	167,000	167,000
adj. R sq.	0.39	0.39	0.31	0.17	0.16	0.27

Note. Difference-in-differences for the weighted annual R&D survey responses for worldwide variables. The dependent variables are total R&D expenses, total R&D performed, total R&D employment, R&D expenses to sales, R&D performed to sales, and R&D employment to total employment, all variables are worldwide measures. Dependent variables in log terms include '+1'. Treatment is the share of *Alice*-treated patents relative to all assigned patents between 2000 and 2009 to firms in the industry, the post-dummy is positive for years after 2014. All observations are weighted by their sample weights assigned by BRDIS, calculated as the ratio of weighted worldwide sales to adjusted worldwide sales. Clustered standard errors are reported in parentheses, $^{*}p<0.1$,$^{**}p<0.05$,$^{***}p<0.01$.

Table 22 Domestic R&D spending.

	Log U.S. R&D Perf.	Log U.S. R&D Emp.	U.S. R&D Perf. to U.S. Sales	U.S. R&D Emp. to U.S. Emp.
Cont. Treatment * Post	-0.204	0.159	0.004	-0.004
	(0.404)	(0.138)	(0.014)	(0.025)
Fixed Effects	Ind. + Year	Ind. + Year	Ind. + Year	Ind. + Year
S.E. Cluster	Ind.	Ind.	Ind.	Ind.
N	167,000	167,000	167,000	167,000
adj. R sq.	0.39	0.31	0.00	0.27

Note. Difference-in-differences for the weighted annual R&D survey responses for domestic/US variables. The dependent variables are total R&D performed, R&D employment, R&D performed to sales, and R&D employment to total employment, all variables are US/domestic measures. The regression sample and specifications are the same as in table 21. Clustered standard errors are reported in parentheses, $^{*}p<0.1$,$^{**}p<0.05$,$^{***}p<0.01$.

BRDIS also asks for the type of R&D spending. In this context, research is related to the exploration of new ideas, and BRDIS distinguished research further into basic research as "activities aimed at acquiring new knowledge or understanding without specific immediate commercial applications" and applied research "aimed at solving a specific problem or meeting a specific commercial objective". Development is more focused on own innovation and the quality of products through the "systematic use of research and practical experience to produce new or significantly improved goods, services, or processes".[46]

In table 23, I show results for the four R&D spending types measured relative to worldwide R&D expenses. The goal is to measure innovation direction rather than overall effort by normalizing with total R&D spending. Development spending has a positive DiD coefficient of 0.26, while research, applied research, and basic research spending all decrease with DiD

[46] See `www.census.gov/programs-surveys/brds/about/faq.html`

Table 23 R&D spending by type.

	Research	Development	Applied Research	Basic Research
Cont. Treatment * Post	-0.321***	0.265***	-0.145*	-0.174**
	(0.082)	(0.087)	(0.080)	(0.077)
Fixed Effects	Ind. + Year	Ind. + Year	Ind. + Year	Ind. + Year
S.E. Cluster	Ind.	Ind.	Ind.	Ind.
N	88,500	88,500	88,500	88,500
adj. R sq.	0.15	0.15	0.10	0.11

Note. Difference-in-differences for the weighted annual R&D survey responses for R&D spending types relative to worldwide R&D expenses. The dependent variables are research spending relative to worldwide R&D expenses, development spending relative to worldwide R&D expenses, applied research spending relative to worldwide R&D expenses, and basic research spending relative to worldwide R&D expenses. Treatment is the share of *Alice*-treated patents relative to all assigned patents between 2000 and 2009 to firms in the industry, the post-dummy is positive for years after 2014. All observations are weighted by the sample weight assigned by BRDIS, calculated as the ratio of weighted worldwide sales to adjusted worldwide sales. Clustered standard errors are reported in parentheses, $^{*}p < 0.1, ^{**}p < 0.05, ^{***}p < 0.01$.

coefficients of -0.32 to -0.15.[47] The overall magnitude of the effects is small: the mean for the treatment variable is 3.1%, thus the average treatment effect for the increase in development spending is 0.8% and for the decrease in research spending ca. 1%. The limited patentability of innovations after *Alice* slightly changes the direction of innovation away from exploration-oriented research toward internal quality improvement-oriented development.[48]

To summarize the industry results, I find a decrease in new establishment creation by large incumbent firms after *Alice* in the most affected industries. Firms are also less likely to receive a patent in the year of entry, providing direct evidence that the limited patentability of inventions is linked to the decrease in new establishment creation. R&D survey responses indicate no decrease in R&D effort but rather a redirection of innovation efforts toward development rather than research. Overall, this is in line with the limited patentability of innovations incentivizing incumbents to focus more on improving current products rather than exploring new ideas, leading to less creative destruction in favor of more own innovation.

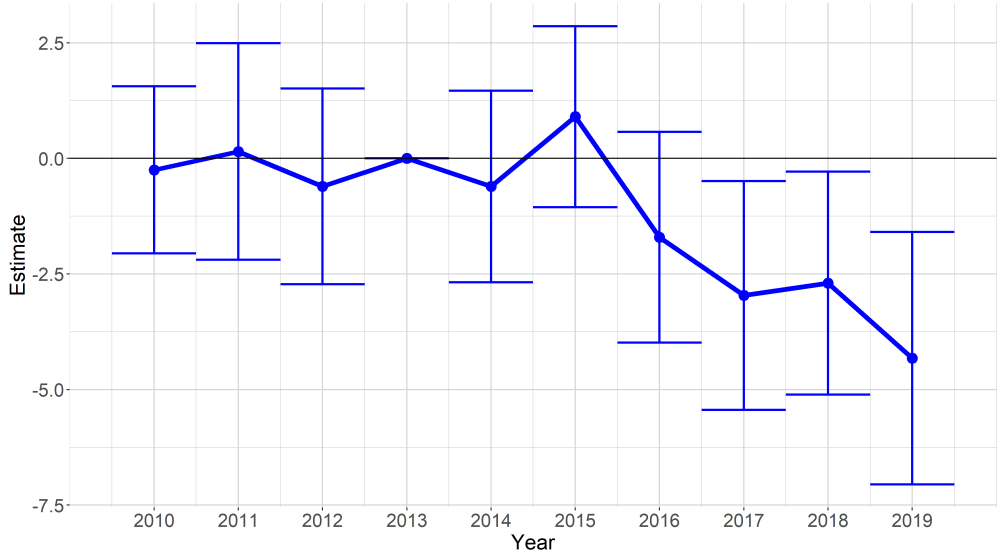
[47] Note that the changes in innovation spending types do not add up to 1 since the survey design does not require that the responses add up to the total R&D spending.

[48] Less research and more development do not mean that *Alice*-treated patents are primarily research related. Patents are relevant for both, research and development, and I use patent measures in section 3.2 to analyze how the innovation direction changes after *Alice*.

Online Appendix C: Event study graphics

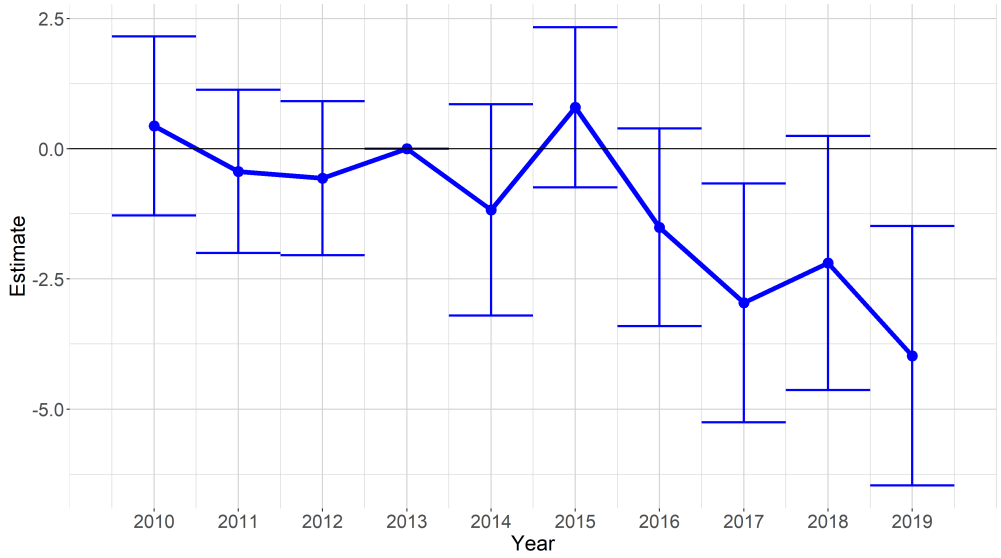
In this section, I present the results of event study analysis using model 2 for different outcome variables. I summarize the results by plotting graphics of the treatment effects using error bars to represent the 95% confidence interval. The models do not contain control variables, but their inclusion does not change the overall results.

Figure 5 Event-study - Entry Emp. - >90th pctl. pat. - >90th pctl. emp.



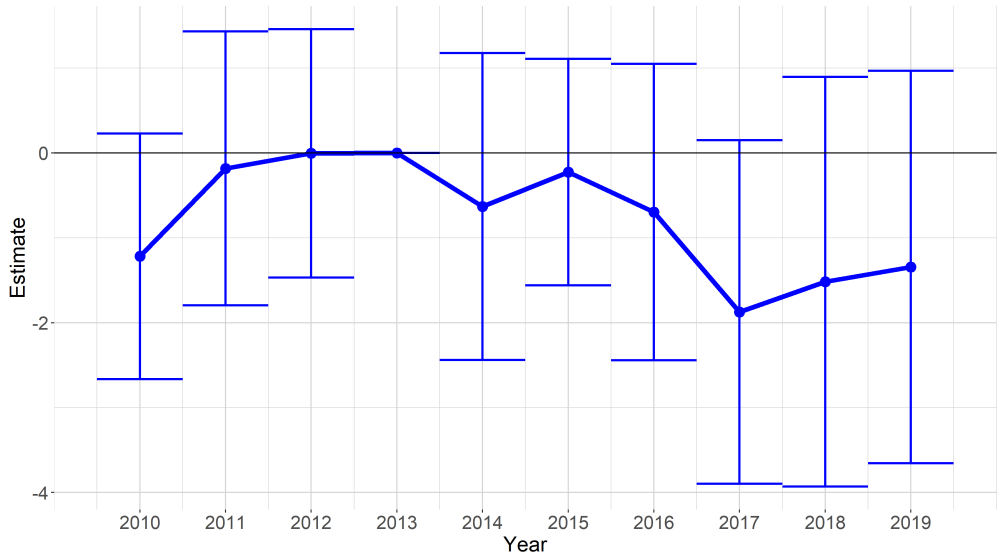
Note. Event study for entry employment for the group of firms above the 90th percentile of the within-industry patent portfolio size and firm employment size distribution in 2009 as defined in table 10. Observations are on the firm-year level. Outcome variables and treatment shares are defined in appendix A. Treatment effects are estimated with the event-study model in equation 2. Dependent variables are in log terms and include '+1'. Coefficients for the interaction of the treatment variable with the year dummies are shown with error bars for the 95% confidence interval. Standard errors are clustered on the firm level.

Figure 6 Job Creation - >90th pctl. pat. - >90th pctl. emp.



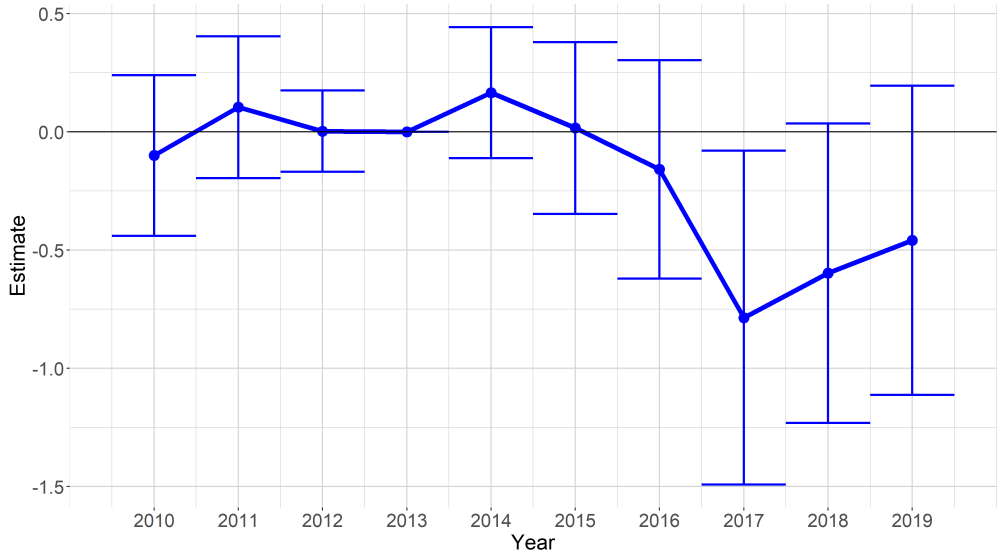
Note. Event study for job creation for the group of firms above the 90th percentile of the within-industry patent portfolio size and firm employment size distribution in 2009 as defined in table 10. Observations are on the firm-year level. Outcome variables and treatment shares are defined in appendix A. Treatment effects are estimated with the event-study model in equation 2. Dependent variables are in log terms and include '+1'. Coefficients for the interaction of the treatment variable with the year dummies are shown with error bars for the 95% confidence interval. Standard errors are clustered on the firm level.

Figure 7 Job Destruction - >90th pctl. pat. - >90th pctl. emp.



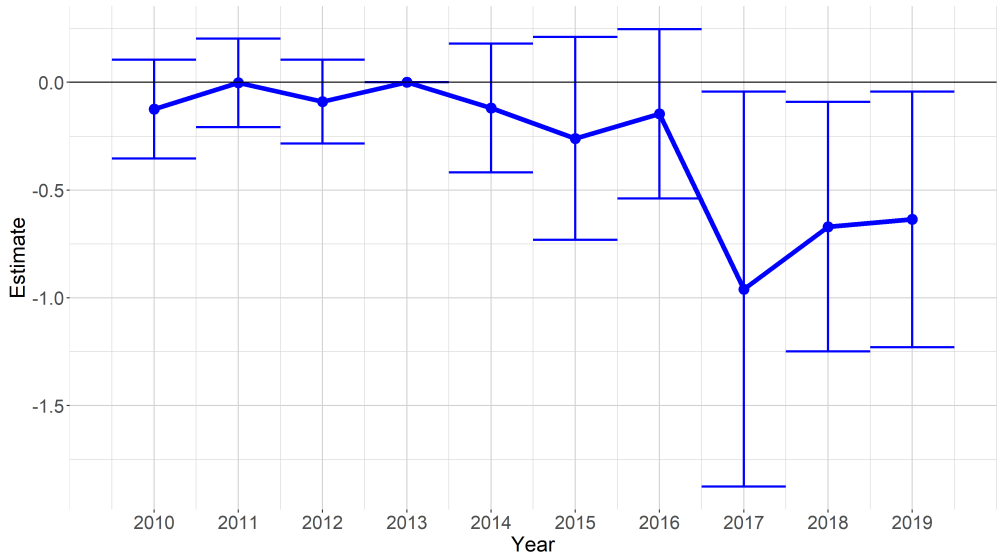
Note. Event study for job destruction for the group of firms above the 90th percentile of the within-industry patent portfolio size and firm employment size distribution in 2009 as defined in table 10. Observations are on the firm-year level. Outcome variables and treatment shares are defined in appendix A. Treatment effects are estimated with the event-study model in equation 2. Dependent variables are in log terms and include '+1'. Coefficients for the interaction of the treatment variable with the year dummies are shown with error bars for the 95% confidence interval. Standard errors are clustered on the firm level.

Figure 8 Establishments (Pos. Emp.) - >90th pctl. pat. - >90th pctl. emp.



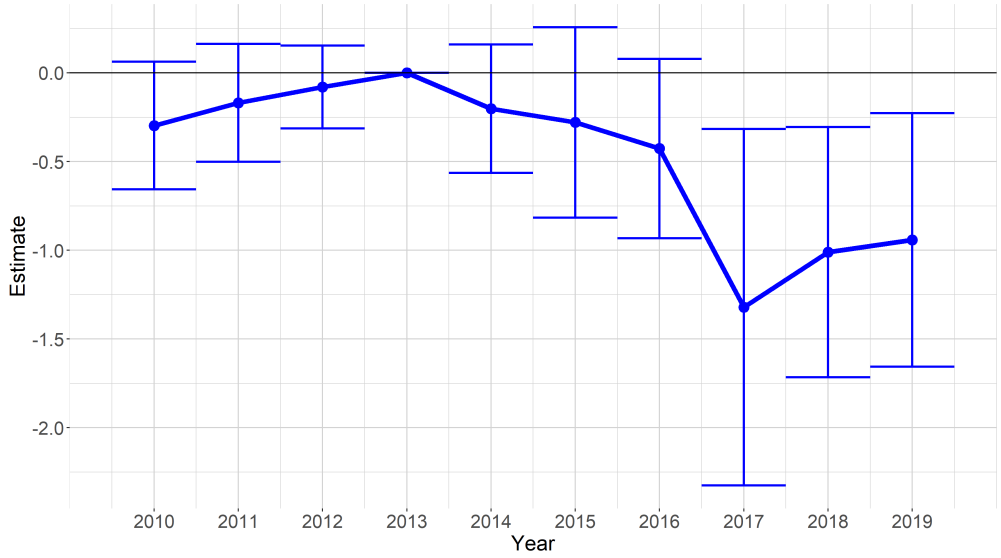
Note. Event study for total establishments with positive employment for the group of firms above the 90th percentile of the within-industry patent portfolio size and firm employment size distribution in 2009 as defined in table 10. Observations are on the firm-year level. Outcome variables and treatment shares are defined in appendix A. Treatment effects are estimated with the event-study model in equation 2. Dependent variables are in log terms. Coefficients for the interaction of the treatment variable with the year dummies are shown with error bars for the 95% confidence interval. Standard errors are clustered on the firm level.

Figure 9 Employment - >90th pctl. pat. - >90th pctl. emp.



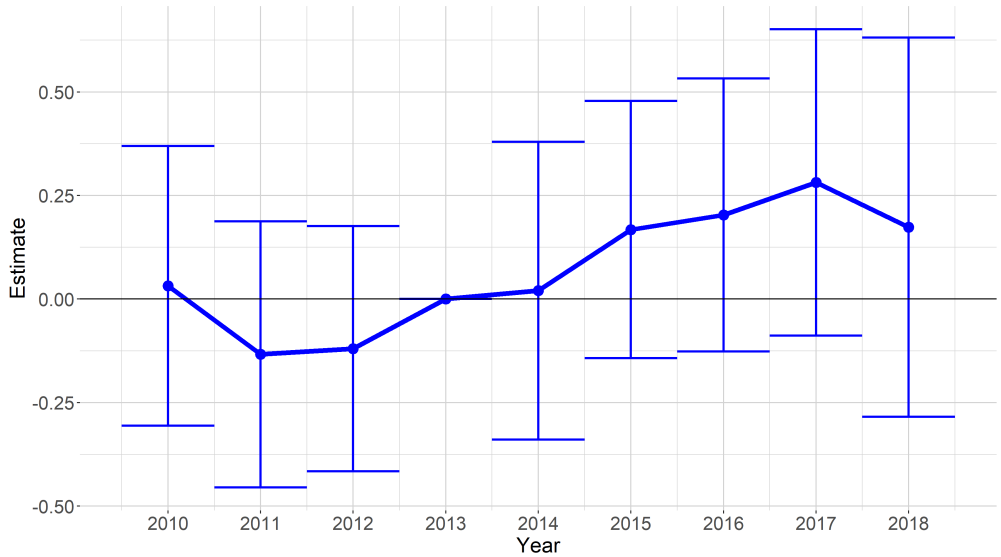
Note. Event study for total employment for the group of firms above the 90th percentile of the within-industry patent portfolio size and firm employment size distribution in 2009 as defined in table 10. Observations are on the firm-year level. Outcome variables and treatment shares are defined in appendix A. Treatment effects are estimated with the event-study model in equation 2. Dependent variables are in log terms. Coefficients for the interaction of the treatment variable with the year dummies are shown with error bars for the 95% confidence interval. Standard errors are clustered on the firm level.

Figure 10 Market Share - >90th pctl. pat. - >90th pctl. emp.



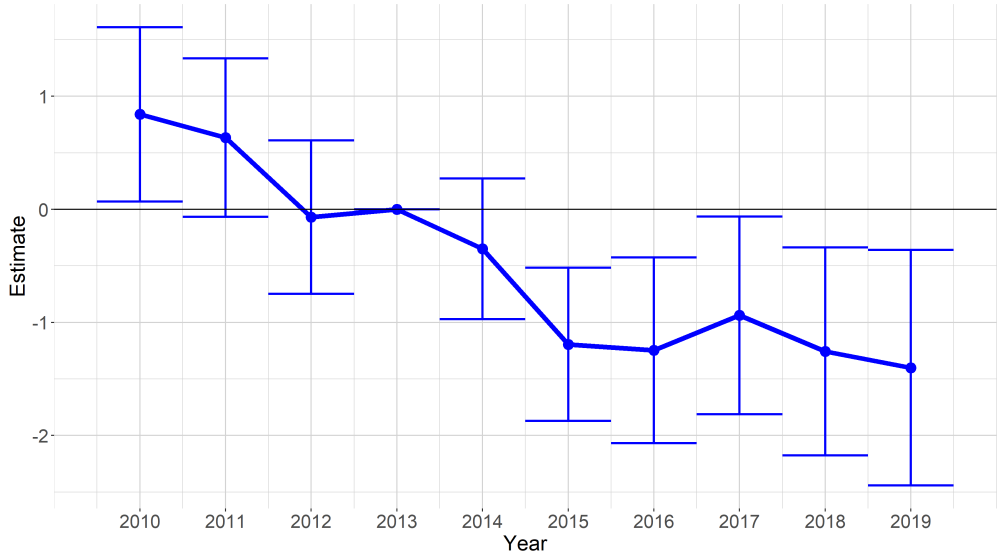
Note. Event study for market share for the group of firms above the 90th percentile of the within-industry patent portfolio size and firm employment size distribution in 2009 as defined in table 10. Observations are on the firm-year level. Outcome variables and treatment shares are defined in appendix A. Treatment effects are estimated with the event-study model in equation 2. Dependent variables are in log terms. Coefficients for the interaction of the treatment variable with the year dummies are shown with error bars for the 95% confidence interval. Standard errors are clustered on the firm level.

Figure 11 Markup - >90th pctl. pat. - >90th pctl. emp.



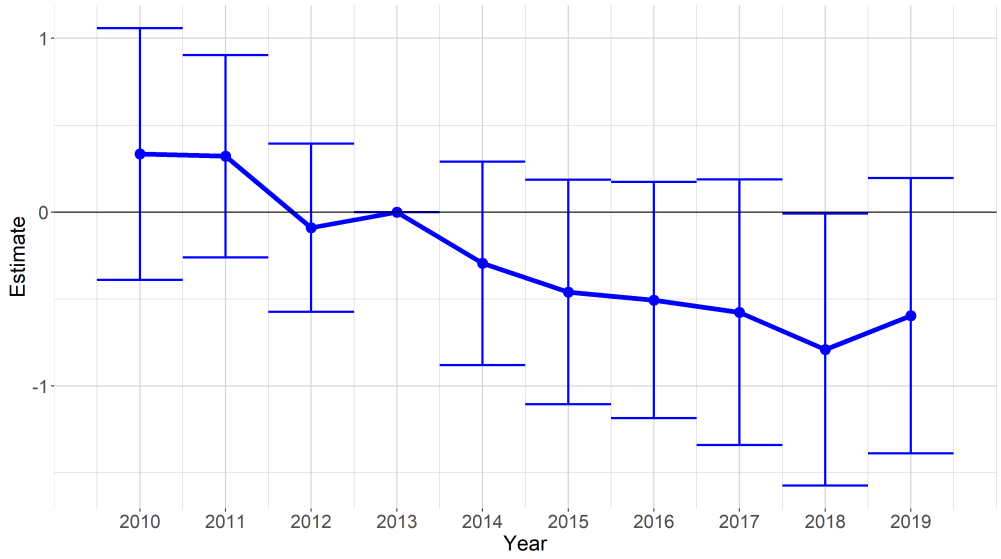
Note. Event study for markups for the group of firms above the 90th percentile of the within-industry patent portfolio size and firm employment size distribution in 2009 as defined in table 10. Observations are on the firm-year level. Outcome variables and treatment shares are defined in appendix A. Treatment effects are estimated with the event-study model in equation 2. Dependent variables are in log terms. Coefficients for the interaction of the treatment variable with the year dummies are shown with error bars for the 95% confidence interval. Standard errors are clustered on the firm level.

Figure 12 Follow-on - >90th pctl. pat. - >90th pctl. emp.



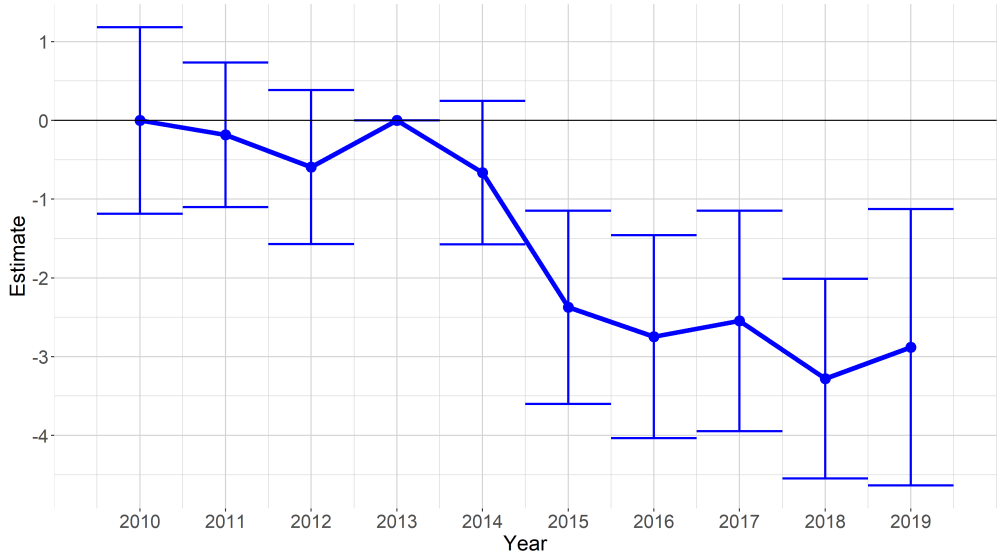
Note. Event study for follow-on citations for the group of firms above the 90th percentile of the within-industry patent portfolio size and firm employment size distribution in 2009 as defined in table 10. Observations are on the firm-year level. Outcome variables and treatment shares are defined in appendix A. Treatment effects are estimated with the event-study model in equation 2. Dependent variables are in log terms and include '+1'. Coefficients for the interaction of the treatment variable with the year dummies are shown with error bars for the 95% confidence interval. Standard errors are clustered on the firm level.

Figure 13 Self-Follow-on - >90th pctl. pat. - >90th pctl. emp.



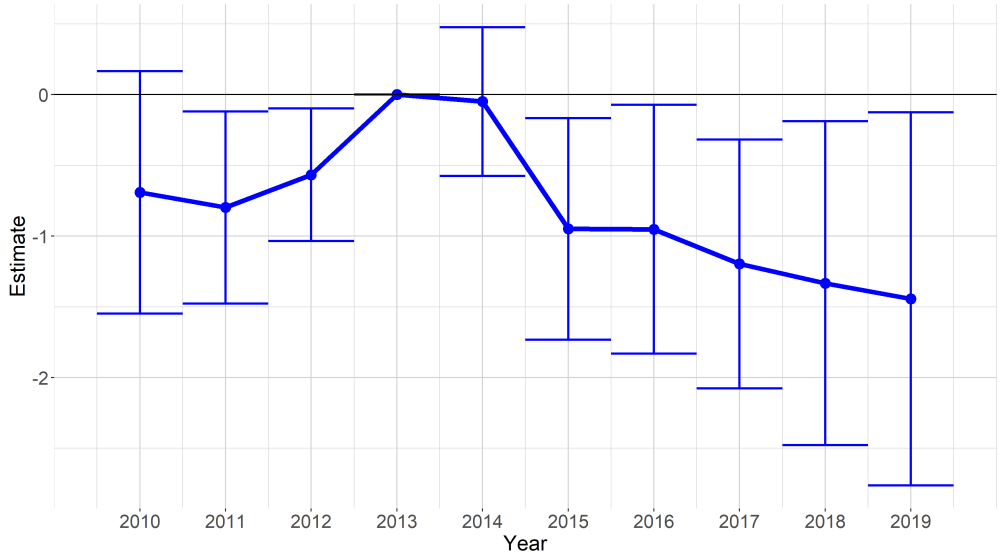
Note. Event study for self-follow-on citations for the group of firms above the 90th percentile of the within-industry patent portfolio size and firm employment size distribution in 2009 as defined in table 10. Observations are on the firm-year level. Outcome variables and treatment shares are defined in appendix A. Treatment effects are estimated with the event-study model in equation 2. Dependent variables are in log terms and include '+1'. Coefficients for the interaction of the treatment variable with the year dummies are shown with error bars for the 95% confidence interval. Standard errors are clustered on the firm level.

Figure 14 Backward Citations - >90th pctl. pat. - >90th pctl. emp.



Note. Event study for backward citations for the group of firms above the 90th percentile of the within-industry patent portfolio size and firm employment size distribution in 2009 as defined in table 10. Observations are on the firm-year level. Outcome variables and treatment shares are defined in appendix A. Treatment effects are estimated with the event-study model in equation 2. Dependent variables are in log terms and include '+1'. Coefficients for the interaction of the treatment variable with the year dummies are shown with error bars for the 95% confidence interval. Standard errors are clustered on the firm level.

Figure 15 Self-Citations - >90th pctl. pat. - >90th pctl. emp.



Note. Event study for self-citations for the group of firms above the 90th percentile of the within-industry patent portfolio size and firm employment size distribution in 2009 as defined in table 10. Observations are on the firm-year level. Outcome variables and treatment shares are defined in appendix A. Treatment effects are estimated with the event-study model in equation 2. Dependent variables are in log terms and include '+1'. Coefficients for the interaction of the treatment variable with the year dummies are shown with error bars for the 95% confidence interval. Standard errors are clustered on the firm level.

Online Appendix D: Model and welfare analysis

I use a simplified Schumpeterian growth model based on Peters (2020) to quantify the welfare effect of the *Alice* decision. I calibrate changes to the innovation rates of technology exploration and own innovation with the DiD results from section 3.5.

The economy has a representative household with log utilities in continuous time. The household consumes the single good C_t with the following preferences:

$$U = \int_0^\infty e^{-rt} \ln C_t dt$$

where r is the risk-neutral interest rate.

C_t is a fixed share of the final output Y_t. Y_t is produced competitively with a unit mass of intermediate inputs according to the following Cobb-Douglas technology:

$$\ln Y_t = \int_0^1 \ln(q_t(i) y_t(i)) di$$

where $y_t(i)$ is the quantity and $q_t(i)$ the quality of intermediate good i.[49]

There are J incumbent firms indexed by j. There is no entry since the results in appendix B show that new firm creation is not affected by *Alice*. Each firm j has the knowledge to produce product i with quality $q(i,j) > 0$ using the following linear technology:

$$y(i,j) = l(i,j)$$

where $l(i,j)$ is labor used by firm j to product product i.

Firms engage in Bertrand competition, which implies that only the quality leader is active in a given market. I denote the index of the leading firm in product i by $j(i)$. Product quality $q(i)$ is a quality ladder with proportional improvements of size $\lambda > 1$. The quality gap between the leader $j(i)$ and the second-highest quality producer is $\lambda^{\Delta(i)}$. That is, firm $j(i)$ is $\Delta(i)$ rungs above the second-highest quality producer.

Under Bertrand competition, the markup $\mu(i)$ that $j(i)$ can charge for product i depends on its quality lead:

[49] I drop in the following the time subscript if there is no ambiguity.

$$\mu(i) = \lambda^{\Delta(i)}$$

Since the household has an elasticity of substitution of one, it spends the same amount on each product i. I normalize the amount spent on each product to one. The flow profits for firm $j(i)$ from product i thus are given as follows:

$$\pi(i) = \left(1 - \frac{1}{\mu(i)}\right) = \left(1 - \frac{1}{\lambda^{\Delta(i)}}\right)$$

Firm $j(i)$ can lose product i through creative destruction if a competitor successfully innovates on product i and becomes the new quality leader. I model creative destruction as the flow rate τ. Since I observe no significant effect of *Alice* on establishment exit or job destruction, I take τ as exogenous.[50]

I can now calculate the present value (PV) $V(i)$ of product i for its quality leader $j(i)$. The discount rate combines the risk-neutral interest rate r and the rate of creative destruction τ. In other words, firm $j(i)$ takes into account the time value of expected future profits and the probability of losing the product to a competitor.

$$V(i) = \int_0^\infty \pi(i) e^{-(r+\tau)s} ds = \frac{\pi(i)}{r+\tau} = \frac{\left(1 - \frac{1}{\mu(i)}\right)}{r+\tau} \tag{3}$$

Firm j can engage in two types of innovation: first, it can improve quality $q(i)$ through own innovation for products in which it is the quality leader and increase the gap to the second-highest quality producer from $\Delta(i)$ to $\Delta(i)+1$. Second, firm j can engage in new technology exploration and expand into a new product k. If expansion innovation is successful and the firm takes over product k, it will take the quality lead only by a single rung above the second-highest quality producer such that $\mu(k) = \lambda$ and the quality gap becomes $\Delta(k) = 1$.

Both types of innovation are random and firm j invests into R&D to decide on the company-wide Poisson arrival rate for own innovation I_j and expansion innovation x_j.[51] I assume that R&D cost and the own innovation and expansion rates are proportional to the

[50] As mentioned in section 3.5, I observe toward the end of my sample period a negative treatment effect for job destruction and establishment exit, which would be consistent with a decrease in τ and a new equilibrium.

[51] In equilibrium, the sum of the expansion innovation efforts across all firms in J would be equal to the overall rate of creative destruction: $\tau = \sum_{j=1}^{J} x_j dj$.

number of products the firm currently owns as quality leader n_j. With a quadratic cost function, I can express the R&D costs for firm j as follows:

$$C(I_j, x_j, n_j) = n_j \frac{\xi_I}{2} I_j^2 + n_j \frac{\xi_x}{2} x_j^2$$

where ξ_I and ξ_x are scale parameters.

I can use equation 3 to find expressions for the optimal own innovation and expansion rates per product by setting the expected marginal increase in $V(i)$ equal to the marginal cost of either innovation effort.

$$\begin{aligned} \frac{\left(1 - \frac{1}{\lambda}\right)}{r + \tau} &= \xi_x x \\ \frac{\left(\frac{1}{\lambda^{\bar{\Delta}}} - \frac{1}{\lambda^{\bar{\Delta}+1}}\right)}{r + \tau} &= \xi_I I \end{aligned}$$

where $\bar{\Delta}$ is the average quality lead of firm j across its current products, and x and I are the optimal innovation rates.

Expanding into a new product is more profitable than improving current products since $1 - \frac{1}{\lambda} > \frac{1}{\lambda^{\bar{\Delta}}} - \frac{1}{\lambda^{\bar{\Delta}+1}}$. Thus, the flow profits $\pi(k)$ from expanding into new product k are larger than the increase in flow profits for successful own innovation on current product i. However, we can assume that expansion innovation is also more costly since firm j needs to acquire more external knowledge and integrate new processes when entering the new market k.

My empirical results in appendix B.1 show no change in overall R&D efforts for treated industries but rather a substitution effect toward product development. My patenting results in section 3.2 also confirm that *Alice* does not prevent breakthrough innovations from being patent eligible but leads to less exploration and more internal product improvement. If I assume that *Alice* makes exploration innovation more costly and that firms have a limited innovation budget, firms substitute the least valuable expansion projects with the next most valuable own innovation projects.

Formally, firm j optimizes the following Lagrangian:

$$\max_{x,I} \left\{ x \frac{\left(1 - \frac{1}{\lambda}\right)}{r + \tau} + I \frac{\left(\frac{1}{\lambda^{\bar{\Delta}}} - \frac{1}{\lambda^{\bar{\Delta}+1}}\right)}{r + \tau} + \Lambda \left(B - \frac{\xi_I}{2} I^2 - \frac{\xi_x}{2} x^2 \right) \right\}$$

where B is the R&D budget and the constraint is binding. We can ignore the scaling by the number of products n_j since innovation costs, innovation rates, and the project values scale with the product count. The solution shows that the ratio of expected profit increases from successful innovation is equal to the ratio of the marginal innovation costs:

$$\frac{\left(1-\frac{1}{\lambda}\right)}{\left(\frac{1}{\lambda^{\Delta}}-\frac{1}{\lambda^{\Delta+1}}\right)}=\frac{\xi_x x}{\xi_I I} \tag{4}$$

In this context, what does *Alice* mean for the decision to invest into own innovation versus expansion innovation? Limited patent protection means that expansion costs increase since entering a new market requires additional efforts by the firm to protect the invention against imitation risk and license external knowledge. In equation 4, increasing ξ_x relative to ξ_I on the right-hand side of the equation predicts the same shift away from expansion innovation and new technology exploration toward own innovation as I find in my empirical results after the *Alice* shock.

D.1. Welfare calibration

I can use the estimated treatment effects from section 3.5 and measure how economic welfare changes when the largest treated firms in software-oriented industries shift their innovation direction away from technology exploration toward own innovation. Since *Alice* is quite sector-specific, I focus on the welfare impact for just the most treated industries.

Following Peters (2020), aggregate output is proportional to the quality index of products, $Q=\exp\left(\int_0^1 \ln q(i)\ di\right)$.[52] Economic growth depends only on the overall quality improvements to the intermediate products:

$$g=\sum_J s_j(x+I)\ln(\lambda)$$

[52] Aggregate labor demand L is the sum of the individual firm-level labor demand: $L=\sum_J l_j=\frac{Y}{w}\sum_J n_j\frac{1}{\mu_j}$. I can use this to find an expression for the real wage w: $\ln Y=\int_0^1\ln\left(q(i)y(i)\right)di=\int_0^1\ln q(i)di+\int_0^1\ln y(i)di=\ln Q+\int_0^1\ln\left(\frac{Y}{w}\frac{1}{\mu(i)}\right)di=\ln Q+\ln Y-\ln w-\int_0^1\ln\mu(i)di$. Thus, the equilibrium wage w is given as $w=Q\exp\left(-\int_0^1\ln\mu(i)di\right)$. I can substitute this into the expression for the aggregate labor demand and define the misallocation measure $M=\frac{\exp\left(-\int_0^1\ln\mu(i)di\right)}{\sum_J n_j\mu_j^{-1}}$ such that $Y=QML$. Since I am interested in changes to growth rather than the full equilibrium, I can focus just on the change in overall quality Q and take L and M as constant.

Table 24 Calibration - welfare analysis.

Parameter	Name	Estimate	Source
r	Risk-neutral interest rate	0.03	Akcigit and Kerr (2018)
λ	Innovation step size	1.3	Hall (2018)
x	Expansion innovation rate	0.08	Pre-period average of Establishment Entry Rate in BDS Data[a]
I	Own innovation rate	0.05	Ration of Self-Follow-on Citations to Follow-on Citations for firms above the 90th percentile of the patent portfolio and employment size distribution in table 10, solved for $\frac{I}{I+x}$
g	Growth rate	0.034	Pre-period average of TFP Growth in BLS Data[b]
s_j	Market share	0.26	Pre-period average for treated firms above the 90th percentile of the patent portfolio and employment size distribution as defined in table 10
d_j	Continuous treatment	0.16	Pre-period average for treated firms above the 90th percentile of the patent portfolio and employment size distribution as defined in table 10
	Treatment effect expansion innovation	-0.6	DiD-coefficient for establishment entry in table 14
	Treatment effect own innovation	+0.2	DiD-coefficient for markup in table 15

Note.

[a] Source: `www.census.gov/data/datasets/time-series/econ/bds/bds-datasets.html`

[b] Source: `www.bls.gov/productivity/tables/major-industry-total-factor-productivity-klems.xlsx`

where s_j is the market share of firm j, x and I are the optimal innovation rates, and λ is the quality ladder step size.[53] For simplicity, I assume here that all firms solve for the same optimal own innovation and exploration innovation rates before *Alice*.

Since Y_t only grows through quality improvements at rate g and $C_t \propto Y_t$, i.e., consumption is a fixed share of total output, the change in welfare following *Alice* is the discounted present value of the (permanent) change in the growth rate $\frac{\Delta g}{r}$.

Table 24 summarizes the parameters I use to calibrate my model. First, I set the risk-neutral interest rate r to 3%, similar to the value used in Akcigit and Kerr (2018). Next, I need an estimate for the innovation step size λ. I follow Aghion et al. (2019) and use the

[53] I define the market share of firm j as the total employment of firm j relative to the sum of total employment across all firms in the four-digit NAICS industry. Note that $\frac{l_j}{L} = \frac{n_j \frac{1}{\mu_j}}{\sum_J n_j \frac{1}{\mu_j}}$; if I assume that markups are relatively evenly distributed across companies, this simplifies to $\frac{l_j}{L} = \frac{n_j}{\sum_J n_j}$.

inverse of the markups estimated in Hall (2018) to calibrate λ.[56] Since table 3 shows that the two most treated sectors are 5415 and 5112, I focus on the markup estimates for Professional, Scientific, and Technical Services (NAICS 54) and Information (NAICS 51) in Hall (2018), which are 1.31 and 1.39, respectively. I round to 1.3 to account for estimation errors.

Next, I can use the establishment entry rate as a proxy for the rate of expansion innovation x, similar to Klenow and Li (2021) and Garcia-Macia et al. (2019). I use industry-wide entry rates for sectors 511 and 541 from the Business Dynamics Statistics (BDS). The average annual establishment entry rates for the pre-period (2010 to 2014) are 7.85% and 12.2%, respectively. I approximate the expansion innovation rate with 8%.[57]

I follow an approach similar to Akcigit and Kerr (2018) and use self-citations to proxy for the own innovation rate I. Since backward citations refer to all prior art and might cite more than the relevant technologies (Kuhn et al. 2020), I use follow-on citations for the calibration. I estimate a share of ca. 40% of self-follow-on citations to total follow-on citations and thus set I equal to 0.05.[58]

With these parameters, the growth rate for the most affected industries before *Alice* is equal to:

$$g = (x + I) \ln \lambda = (0.08 + 0.05) * \ln 1.3 = 0.034$$

Compare this to the average total factor productivity (TFP) growth rate estimates for the pre-period from the Bureau of Labor Statistics (BLS): 2.68% for industry 511 and 3.42% for 5415. Thus, the calibrated model leads to realistic estimates close to the observable TFP growth rates for the most treated industries.

I focus on the treatment effects from section 3.5 for large incumbent firms above the 90th percentile of the patent portfolio and employment size distribution to estimate the welfare

[56] Directly using the inverse of the wage share from my data as an estimate for the markup would overestimate the actual innovation step size since large incumbents with successful own innovation in the past have higher markups than the simple inverse of the step size. The wage share for the largest patent portfolio/employment size group in the summary statistics in table 10 is 0.23 and for all other size groups ca. 0.3. If I calibrate λ to 1.3, the ratio of the inverse of the wage share for the firms in the largest patent portfolio/employment size group to the rest indicates an average innovation lead of $\Delta' = 2$ for the largest firms; $\frac{\mu'}{\mu} = \frac{\lambda^{\Delta'}}{\lambda^{\Delta}} = \frac{\frac{1}{0.23}}{\frac{1}{0.3}} = 1.3 = \lambda^{2-1}$.

[57] Note that the establishment entry rates lie between the employment-weighted entry rates (3.82% for 511 and 5.66% for 541) and the job creation rates (11.2% for 511 and 16.3% for 541).

[58] In this calibration, patent citations are proportional to the innovation rates. The share of self-follow-on citations is equal to $\frac{I}{I+x} = 0.4$, thus $I = \frac{2}{3} * x = 0.05$.

impact of *Alice*. The average market shares of these large treated firms in sectors 51 and 54 are 0.37 and 0.15, respectively, and the averages of the continuous treatment d_j are 0.22 and 0.10. Thus, there is a relatively wide distribution of market shares and treatment across industries. For this analysis, I take the mid-points for either variable, setting the market share of treated firms to 0.26 and the average treatment to 0.16.

I use the treatment effects estimated in tables 14 and 15 of -0.6 for establishment entry and +0.2 for markups to measure the decrease in expansion innovation and the increase in own innovation for treated firms after *Alice*.[59]

I multiply the treatment effect for establishment entry with the calibrated pre-period expansion innovation rate to calculate the effect of *Alice* on expansion innovation. Since I focus on the largest treated firms, I weigh the effect with the average continuous treatment variable and the market share of the firms:

$$\Delta x = (-0.6 * 0.08) * (0.26 * 0.16) = -0.0020$$

Similarly, I can estimate the increase in own innovation using the treatment effect for markups and the pre-period own innovation rate:

$$\Delta I = (+0.2 * 0.05) * (0.26 * 0.16) = 0.0004$$

The net change to overall growth combines the two effects and the innovation step size:

$$\Delta g = (\Delta x + \Delta I) \ln \lambda = (-0.0020 + 0.0004) * \ln 1.3 = -0.0004$$

The total welfare loss is the present value of this permanent decrease in consumption growth:

$$\frac{\Delta g}{r} = \frac{-0.0004}{0.03} = -0.013$$

[59] The treatment effects for entry employment and job creation are much larger, between -1.15 and -1.27 for the specification with covariates interacted with year dummies. Thus, the treatment effect for establishment entry is a conservative estimate for the welfare impact of *Alice*.

Thus, *Alice* leads to a decrease in welfare of ca. 1.3% across the most treated industries. The welfare effect of *Alice* is relatively moderate since large firms change the innovation direction toward more own innovation rather than decrease overall innovation efforts.

I can also use the *Alice* shock to estimate the elasticity of growth to patentability, i.e., by how much growth changes if the patent protection of inventions changes. For my calibrated model, the decrease in the growth rates is equal to the following:

$$\frac{\Delta g}{g} = \frac{-0.0004}{0.034} = -1.18\%$$

I can take the decrease in total patent issuances as a proxy for the decrease in patent protection of innovations since the overall innovation effort seems to not have changed after *Alice*. Table 13 shows a treatment effect for total patent issuances of ca. -1.4. Since the dependent variable is in log terms, I only need to weight the treatment effect with the market share of treated firms of 0.26 and the average treatment of 0.16 from above to estimate the percentage change in patent issuances:

$$\frac{\Delta \text{Patent Issue}}{\text{Patent Issue}} = -1.4 * (0.26 * 0.16) = -5.82\%$$

This implies an elasticity of $-1.18/-5.82 = 0.2$ for the relationship of growth to patentability. A 1% decrease in the patentability of innovations leads to a decrease in growth of 0.2% due to large firms shifting from new technology exploration toward own innovation.

Online Appendix E: NLP methodology

I use natural language processing (NLP) to analyze the language of patents and define the intellectual property portfolio held by firms and their exposure to the *Alice* Supreme Court decision. I first develop and train a model to identify claim language that is invalid under *Alice*. Second, I classify existing patents and create annual patent portfolios using patent assignments to firms. Finally, I validate my classification using data on lawsuits and find that my NLP-based classification predicts litigated patents after *Alice* better than the patent classes alone.

E.1. NLP approach for treatment identification

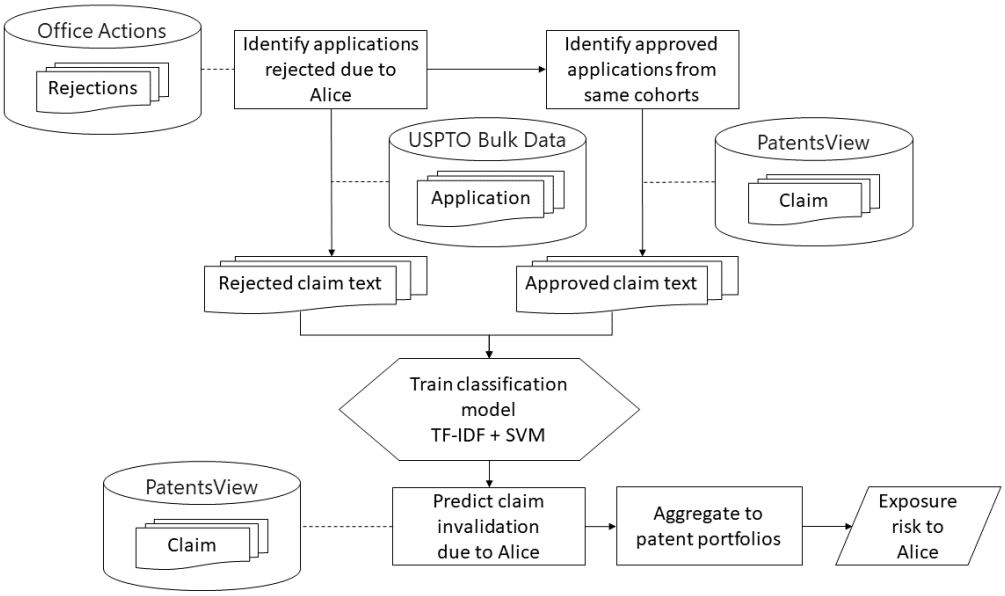
E.1.1. Overview I use rejected patent applications by the United States Patent and Trademark Office (USPTO) mentioning the *Alice* decision as reason to find examples of claim text formulations that are not eligible anymore after 2014. I select approved applications from the same filing years and patent classes as the rejected claims to train a binary classification model on the respective wording. I use this trained classification model on issued patents to predict their exposure to potential invalidation due to *Alice*. Finally, I aggregate the patents that were assigned to firms to patent portfolios and define treatment as the share of invalid patents after *Alice*. Figure 16 illustrates the building and training of the model, the classification process of patent claims, and the definition of firm-wide *Alice* exposure in patent portfolios.

E.1.2. Data and training sets As a first step, I need to identify text examples of claims that are affected by the *Alice* decision. For this, I use rejected patent application data provided by the USTPO and build on the work by Lu et al. (2017).[60]

In this paper, the authors use natural language processing tools on "office actions" to systematically extract and classify the reasons for any rejections, objections, or requirements. An office action is the written notification to the applicant of the examiner's decision on patentability. Since these office actions include references that the applicant may find useful for responding to the examiner and deciding whether to continue pursuing the application, Lu et al. (2017) can identify rejections due to the Supreme Court decision in *Alice Corp. v. CLS Bank International*. The authors restrict to published applications, thus I can use

[60] The data are sourced from the Office of the Chief Economist (OCE), URL: `www.uspto.gov/learning-and-resources/electronic-data-products/office-action-research-dataset-patents`.

Figure 16 Process diagram patent claim classification.



the Patent Examination Research Dataset (PatEx)[61] and published applications[62] to extract the respective independent claim texts and additional application details such as filing date, patent class, and application status.[63] I use applications in the four USPC patent classes with the most rejections due to *Alice*, accounting for more than 91% of all *Alice*-based claim rejections in my sample.[64] Treated claims are those rejected claims that had no other issue identified in the office action and were either final rejections, were abandoned due to failure

[61] Available from USPTO, URL: `www.uspto.gov/learning-and-resources/electronic-data-products/patent-examination-research-dataset-public-pair`.

[62] Applications received since November 2000 are generally published within 18 months following the American Inventors Protection Act (AIPA) (Graham et al. 2015). XML files with weekly pre-grant publications of applications are available under `bulkdata.uspto.gov/`.

[63] I can find for more than 90% of all applications with *Alice* rejections the respective claim publications. If applicants only file in the U.S. and don't seek protection in other national jurisdictions, however, they can request non-publication of an application, which might introduce selection bias into my observable treated claims. Overall, only 3.4% of all *Alice*-treated applications do not have a recorded pre-grant publication in PatEx (making up around 36.9% of all missing observations). Furthermore, looking at the set of applications for which I am not able to find the published pre-grant publications more generally, I do not find evidence for any selection: the time between filing and recorded pre-grant publication is on average 317 days for the applications with texts and 284 days for the missing observations. The likelihoods of being ultimately granted are 57% and 50%, respectively. For the granted applications, the average time between filing and patent issuance is statistically not different between the two groups, with on average 1,519 days for the applications with texts and 1,474 days for the missing observations. Overall, the applications to which I cannot link claim texts are very similar to the applications with texts in terms of USPC classes, filing years, rejection rates, and processing time, suggesting that the missing claim publications are due to issues like typos and errors in the underlying XML files rather than applicants systematically deferring the publication of *Alice*-affected applications.

[64] Those classes are: 705 (Data processing: financial, business practice, management, or cost/price determination), 463 (Amusement devices: games), 702 (Data processing: measuring, calibrating, or testing), 434 (Education and demonstration); with the 705 class accounting for 82.4% of all rejected claims, followed by class 463 with 9.9%, class 702 with 4.6%, and class 434 with 3.1%, each class containing at least 100 rejected claims.

to respond to the office action, or were abandoned after the examiner's answer or board of appeals decision. In total, this results in 5,125 unique treated claim texts.

For the classification algorithm to be trained, I need to have a control group of claim texts that are valid under *Alice*. I select applications from the same USPC classes and the same filing years as the treated applications. I limit to applications that were granted without office action due to 35 U.S.C. §101, that is without issues referring to subject matter eligibility that could be related to the *Alice* decision.[65] I restrict to utility patents issued after the *Alice* decision in June 2014 and use the final claim text from PatentsView; I can be reasonably certain that these issued claim texts were deemed eligible under *Alice* by the patent examiners. To balance my training sets, I randomly draw independent claims as controls, proportional to the distribution of USPC classes and filing years as in my treated claims set.

Figure 17 Word frequency in Alice-affected claims.



Figures 17 and 18 show word clouds of the most frequent words in the treated and control claim corpora, with more frequent words being larger. The clouds look quite similar, which is to be expected for claims from the same patent classes and the same cohorts. The terms are weighted by raw term frequencies without any pre-selection of terms, thus some of the most prominent terms in both groups are terms with little individual meaning such as 'plurality', 'least', and 'wherein', reflecting the legal nature of the claim formulations.

[65] Title 35 of the United States Code (U.S.C.), Section 101 governs which inventions are eligible for patents and reads: "Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title." The *Alice* decision significantly limited the scope of eligible subject matter under the law for certain software and business method inventions that were deemed patent ineligible as a mere computer-based implementation of abstract ideas.

Figure 18 Word frequency in control claims.

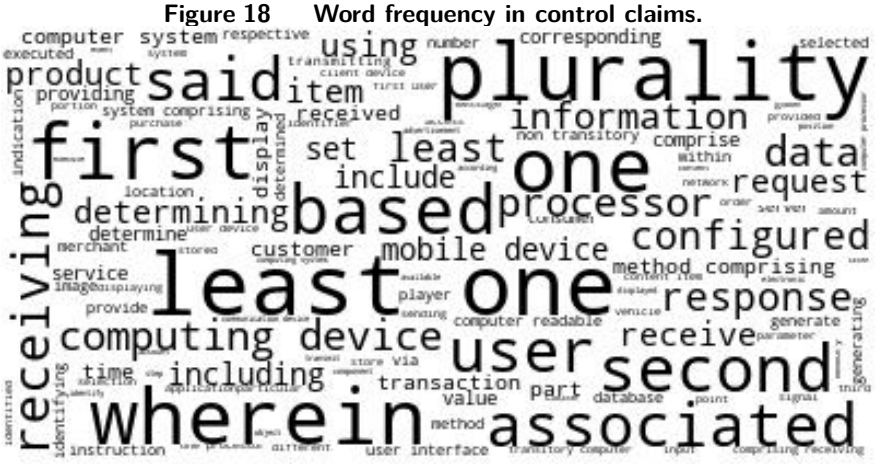


Figure 19 Relative word frequency in Alice-affected claims.

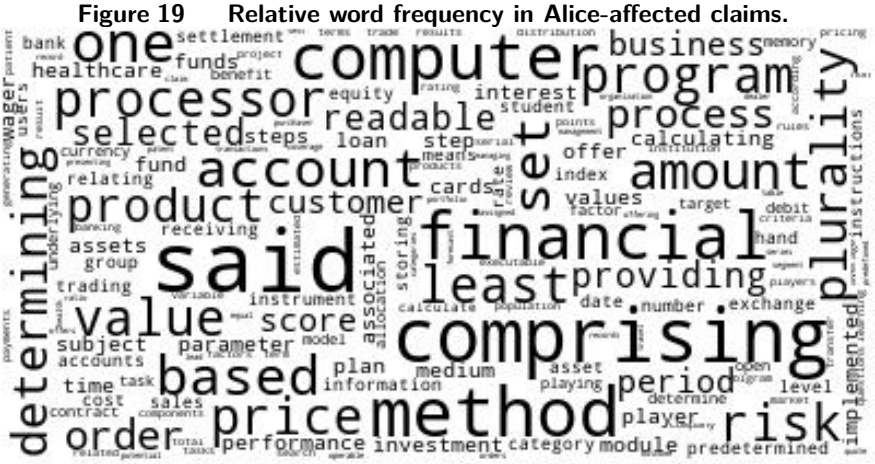


Figure 20 Relative word frequency in control claims.



Figures 19 and 20 show word clouds by weighting terms based on the relative frequency differences, i.e., how more frequent a word is in one group compared to the other. Words like ‘amount’, ‘financial’, ‘risk’, and ‘account’ are more prominent for *Alice*-affected claims, since

the invalidated patents in the case were about an electronic escrow service for facilitating financial transactions. While the eligible claims contain more diverse words including specifications of physical implementations like 'device', 'server', 'display', *Alice*-affected claims prominently feature abstract descriptions about business processes such as 'computer', 'determining', 'method', and 'program'. This is in line with the Supreme Court ruling stating that abstract claims are not eligible just by introducing a computer into the claim.[66] The wording of the treated and control claims thus have distinct characteristics which are not merely explained by patent class or filing cohort but by their exposure to the *Alice* decision.

E.1.3. Model building and evaluation I construct a binary classification model and train it on the rejected claims with the eligible claims as control set to classify patent claims based on their exposure to the *Alice* decision.[67]

Since most classification algorithms require numerical features as input, I vectorize claim texts into arrays representing the words of the claims. I split the text of the claims into separate words (tokens), and remove all numbers, special characters, HTML tags, punctuation, and words shorter than three characters. I also remove stop words, i.e., common words that occur with high frequency but carry little substantive information like 'the', 'and', 'like'. Finally, I convert the resulting strings to lowercase and apply the Porter stemming algorithm. Reducing words to their stems allows us to identify families with similar meanings, e.g., the words 'playing', 'played', and 'play' all have the same stem 'plai'.[68]

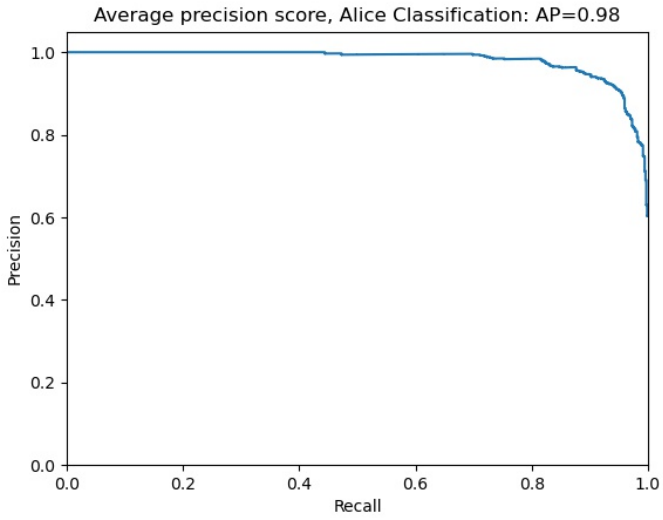
To turn the resulting arrays of tokens into a numeric vector representation, I count the occurrences of each token found in the corpus for each document. I use here uni- and bigrams, i.e., sequences of up to two tokens as features. A single claim is represented as a vector with each element representing the number of occurrences of a token, i.e., the term frequency. The collection of claims thus can be represented by a matrix with one row per claim and one column per token occurring in the corpus. Since tokens that are very common among claims have little value in classifying them, I weigh the term frequency with the inverse

[66] See the Syllabus to *Alice Corporation v. CLS Bank International*, Supreme Court, No. 13–298.: " "Simply appending conventional steps, specified at a high level of generality," to a method already "well known in the art" is not "enough" to supply the " 'inventive concept' " needed to make this transformation [from an abstract idea into a patent-eligible invention]. [...]. The introduction of a computer into the claims does not alter the analysis. Neither stating an abstract idea "while adding the words 'apply it,' " ... is enough for patent eligibility."

[67] All calculations and executions were performed in Python 3.7.8 on a GNU/Linux operating system.

[68] All these pre-processing steps are performed using the 'gensim' library in Python 3 with the function 'preprocess_string'.

Figure 21 Precision-recall curve.



document frequency, i.e., the frequency of the term in the overall corpus. The more common a term is across all the claims, the lower the weight of the respective column in the weighted term frequency matrix.[69] Overall, the texts of treated and control claims are represented as numeric vectors in a Term Frequency–Inverse Document Frequency (TF-IDF) matrix, giving more weight to the unique language of claims.

I train a support vector machine and construct a hyperplane to separate eligible from ineligible claims in the TF-IDF matrix. I use a quadratic kernel to solve for non-linear relations.[70] The full model thus takes as input the full text of a claim and gives a predicted probability between 0 and 1 of whether this claim was affected by the *Alice* decision. I use 85% of the training data for model building and the rest to evaluate the quality of the fit. Table 25 reports the performance of my classification model for this test dataset, figures 23, 21, and 22 give a graphical interpretation with the confusion matrix, precision-recall curve, and receiver operating characteristic (ROC) curve.

The confusion matrix plots the relative share of correctly predicted labels in the diagonal from the upper left to the bottom right corner. Related measures are precision, defined as the number of true positives over the number of true positives plus the number of false positives, and recall, defined as the number of true positives over the sum of true positives and false

[69] The specific inverse document frequency weight for feature t is $idf(t) = \ln \frac{1+n}{1+df(t)} + 1$ with n being the total number of claims in the training set and $df(t)$ the number of claims in the corpus containing the term t. The weighted vectors are normalized to unit length.

[70] Using a linear or RBF kernel does not change the results. I use the 'scikit-learn' module in Python to implement the TF-IDF matrix and the support vector classification (SVC).

Figure 22 ROC curve.

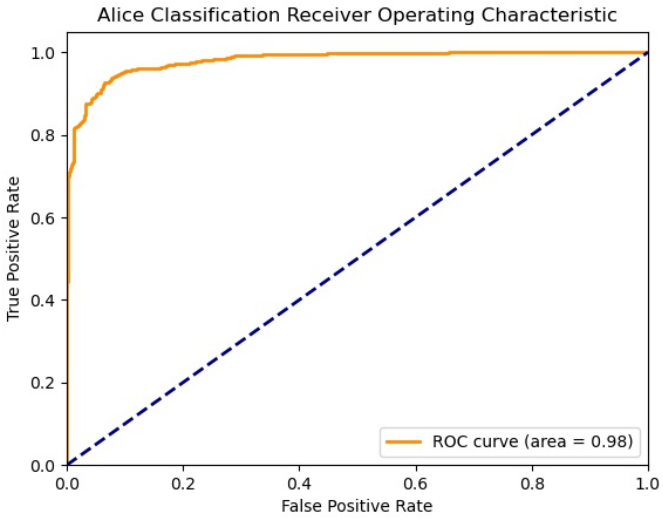
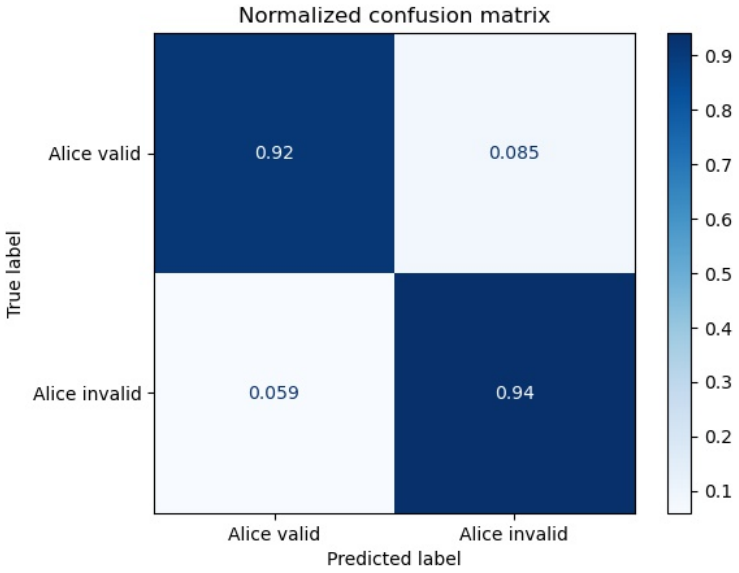


Figure 23 Confusion matrix.



negatives. I want to achieve a high precision, i.e., my model correctly classifies claims as affected, and a high recall, i.e., my model finds most instances of *Alice*-affected claims.

There is a trade-off between these two measures since higher thresholds for being classified as affected return fewer results, but most of the predicted labels are correct (i.e., high precision). On the other hand, if the threshold is very low, the model returns many results, i.e., has a high recall, but most of its predicted labels are incorrect. The precision-recall curve plots this trade-off for different threshold levels with a larger area under the curve indicating a better classification model. The ROC curve similarly plots the fraction of true positives out

Table 25 Classification report for NLP model.

	Precision	Recall	F1 Score	Support
Valid Claims	0.940	0.915	0.927	696
Invalid Claims	0.917	0.941	0.929	696
Precision Score	0.928			
Recall Score	0.928			
Accuracy Score	0.928			
F1 Score	0.928			
MCC	0.857			

Table 26 Comparison of different NLP models.

Model Name	Precision Score	Recall Score	F1 Score	Accuracy Score	MCC
AdaBoost	0.724	0.739	0.731	0.728	0.457
Decision Tree	0.746	0.777	0.761	0.756	0.513
Gradient Boosting	0.767	0.792	0.779	0.776	0.552
K-nearest Neighbors	0.769	0.773	0.771	0.770	0.540
Logistic Regression	0.813	0.829	0.821	0.819	0.638
Naïve Bayes	0.802	0.819	0.810	0.808	0.617
Random Forest	0.876	0.905	0.890	0.889	0.778
Stochastic Gradient Descent	0.852	0.869	0.861	0.859	0.719
Support Vector Machine	0.893	0.935	0.914	0.912	0.824

of the invalid claims against the fraction of false positives out of the valid claims at various threshold settings. The 45-degree line in the ROC plot would represent random guessing while a good model would be in the upper left corner with a high true positive rate and a low false positive rate.

All relevant performance measures are shown in the classification report in table 25. In the lower panel of the classification report, I also report the overall precision and recall scores as well as the accuracy score, which is the average of correctly classified labels. A common measure combining precision and recall is the F1 score, defined as $F1 = 2\frac{precision*recall}{precision+recall}$. Furthermore, the Matthews Correlation Coefficient (MCC) is frequently used to measure the quality of binary classification models.

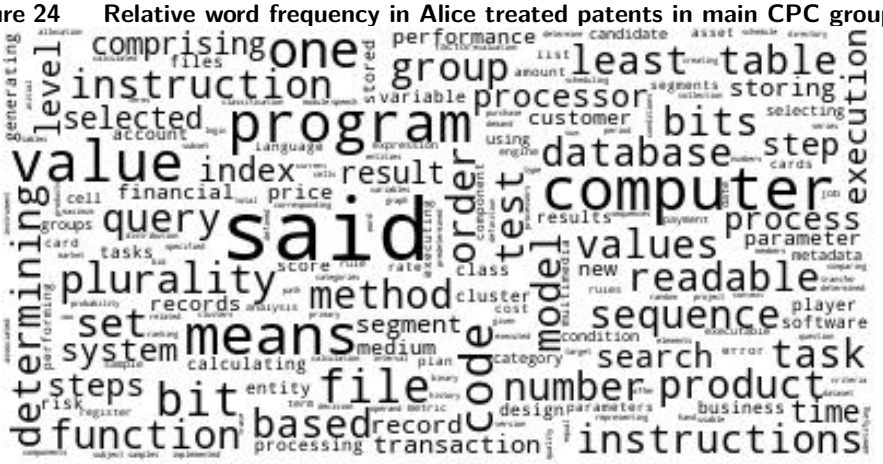
Overall, all measures show a very good classification model with high precision, high recall, and more than 85% MCC correlation, meaning a very high correlation between the true and predicted labels.

Finally, I use the different performance metrics to compare popular classification models on the test data in table 26 to validate my model choice. All models are used with their default parameters and with the same TF-IDF matrix of claim texts. Among all classifiers, the support vector machine proves to be the best.

E.2. Patent classification

Using the trained classification model, I predict the probability of being affected by *Alice* for independent patent claims for issued patents in the treated USPC classes. I do this for all patents issued since 1990 using the claims data from PatentsView.[71] This renders a total of 393,100 classified claims in 115,630 unique patents. Since the USPTO moved to CPC classification in 2013, I repeat the classification process for the five CPC groups most consistent with the treated USPC classes.[72] Classifying patent claims based on CPC groups gives 3,236,420 classified independent claims in 983,229 patents. This is notably more than using the USPC classification due to the USPC classification ending in 2015 in PatentsView data and the affected CPC groups being much broader than the USPC classes.[73] This broader definition of treated patent classes under CPC can also be seen in figures 24, 25, 26, and 27 where I generate word clouds from the claim texts that are predicted to be treated and untreated under *Alice* for the CPC and USPC classes: the CPC group clouds are in general more diverse, though the overall image is similar with words like 'computer', 'means', 'value', 'device', 'signal', and 'display' showing up prominently in both classes and groups.[74]

Figure 24 Relative word frequency in Alice treated patents in main CPC groups.



[71] URL: `patentsview.org/download/claims`.

[72] There is no direct mapping from the USPC classes into CPC groups; however, the groups A63F (Video games), G07F (coin-freed or like apparatus), G06F (Digital data processing), H04L (Transmission of digital information), and G06Q (Data processing systems) represent around 50% of the affected patents in the treated main USPC classes 705, 463, 434, and 702.

[73] The four treated USPC classes account for a total of around 1.1% of all USPC class assignments, while the five corresponding CPC groups make up around 9.2% of all CPC assignments.

[74] I restrict to a sample of 1 million claims from the patents in the classified CPC groups to limit memory usage for the word cloud rendering.

Figure 25 Relative word frequency in Alice treated patents in main USPC classes.

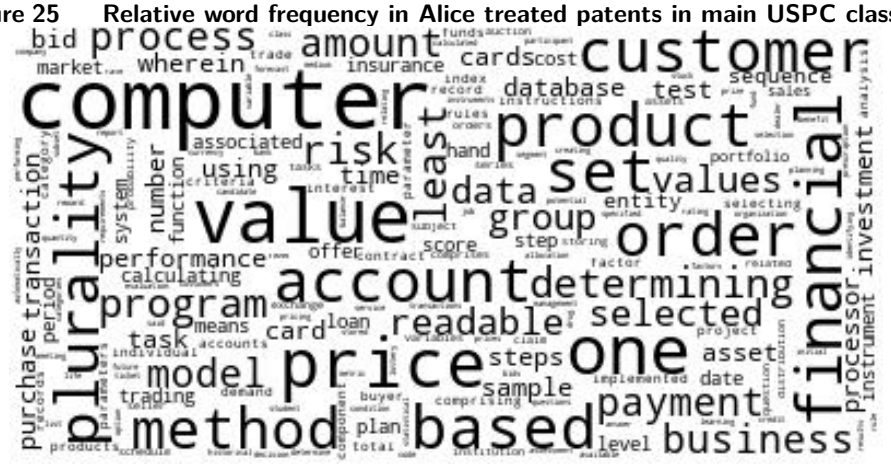


Figure 26 Relative word frequency in untreated patents in main CPC groups.



Figure 27 Relative word frequency in untreated patents in main USPC classes.



To understand better which words are most important for the classification decision, I use a Local Interpretable Model-agnostic Explanations (LIME) algorithm on 1000 randomly selected patent claims. The basic idea behind the LIME algorithm is to perturb the underly-

ing text by leaving out words and measuring how much and in which direction this changes the classification. Doing this gives for each tested text a list of words with the highest positive or negative impact on the classification. In table 27, I show the 20 most frequent words associated with a classification as eligible and ineligible. The general picture from the word clouds of figures 19 and 20 is confirmed: more precise wording with references to the actual implementation of claims improves the eligibility. More abstract and general formulations of methods and processes render claims ineligible. This is consistent with the findings of Dugan (2018) and the literature stating that more specific formulations in claims are eligible under *Alice* compared to abstract ideas and processes (DiNizo 2018, Tran and Benevento 2019, Craig 2017). Notably, the differences between important words for patents in affected CPC groups and USPC classes are small, confirming that the classification model correctly predicts *Alice*-affected patents in the larger CPC groups.

Table 27 LIME important words for classification in 1000 claims sample.

USPC classes		CPC groups	
Ineligible words	Eligible words	Ineligible words	Eligible words
said	second	said	second
means	device	method	device
method	data	means	data
value	signal	value	signal
steps	plurality	program	control
program	display	memory	plurality
player	game	comprising	network
function	control	steps	user
processor	signals	file	information
step	user	group	display
processing	sensor	module	request
condition	information	processing	signals
module	position	processor	application
transaction	server	code	content
comprising	image	level	server
reference	time	sequence	image
level	network	bit	circuit
number	electronic	bits	node
selected	application	input	address
test	circuit	logic	virtual

The words with the highest impact on the classification are ‘said’ for ineligible texts and ‘second’ for eligible texts. These words also show up in the respective word clouds in figures 19, 20, 25, 24, 27, and 26. ‘Said’ and ‘second’ are examples of sentence structures in eligible

and ineligible claims that cannot be captured by, e.g., keyword search alone: while 'said' frequently refers back to a broader subject at the beginning of a claim, 'second' is common in sentences about how several components of an invention are interacting, thus giving a more technical description of the operation. My trained NLP model can uncover and leverage formulations in claim texts like these to more precisely predict whether a claim is eligible or ineligible under *Alice*.

Since most patents granted after 2013 have a CPC classification rather than a USPC classification, I focus on the CPC-based patent classification in the following. I define a patent as being treated if its first claim is predicted to be invalid under *Alice*. The first claim is the broadest claim of a patent and thus sets the scope for the technology protected by the patent (see, e.g., Kuhn and Thompson 2019). Alternative specifications of treatment are as follows: a patent is treated if the majority of its independent claims are predicted to be *Alice*-treated, the average (continuous) predicted treatment probability for all independent claims is above 50%, or at least one independent claim is predicted to be *Alice*-treated. The different treatment definitions on the patent level are highly correlated with correlation coefficients between 71.1% and 88.0%.[75] The only effective difference is in how many patents are predicted to be treated: of the 1,062,883 classified patents, 18.5% are predicted to be treated using the first claim. Using the average treatment probability of claims leads to the fewest patents being predicted to be affected by *Alice* (14.4% of classified patents) and defining treatment if at least one claim is predicted to be ineligible to the most treated patents (24.9% of classified patents). For any of the definitions, the number of treated patents is large with more than 150,000 in each case. In practice, the different definitions of treated patents have little impact on the economic analysis, thus I use in the main analysis the treatment status of the first claim to define treatment on the patent level.

E.3. Patent litigation - validation analysis

I use the Stanford NPE Litigation Dataset, based on Miller et al. (2018), to assess if my classification mechanism can predict real-world patent challenges. The Stanford NPE Litigation Dataset is the "first ever publicly available database to track comprehensively how practicing entities, non-practicing entities, and patent assertion entities (PAEs) claim patent ownership

[75] Patent classification with the main definition using the treatment status of the first claim has a correlation coefficient of 82.6% with the patent classification defining treatment if at least one claim is treated and 88.0% with the definition of treatment if most claims are treated, thus the first claim is representative for the treatment status of the rest of the patent.

rights in litigation. [...] [R]esearchers are tracking every lawsuit filed in U.S. district courts from 2000 to the present and identifying each patent plaintiff as either a practicing entity or as one of eleven types of NPEs."[76] Since confounding legal changes around 2014/2015 have limited the ability for NPEs to file lawsuits (Appel et al. 2019), I focus on lawsuits with product companies as plaintiff. Product companies manufacture products, sell products, or deliver services generally, or are IP enforcement subsidiaries of practicing entities. This category account for around 52.4% of all patent litigation cases in the database. Lawsuits by product companies are also less likely to be 'opportunistic' (Cohen et al. 2019) and the patents defended are related to actual products and used technologies. Thus, if I find that patents predicted to be treated by the NLP method are less likely to be litigated by product companies after 2014, I can confirm that my algorithm identifies patents that are less enforceable after *Alice*.

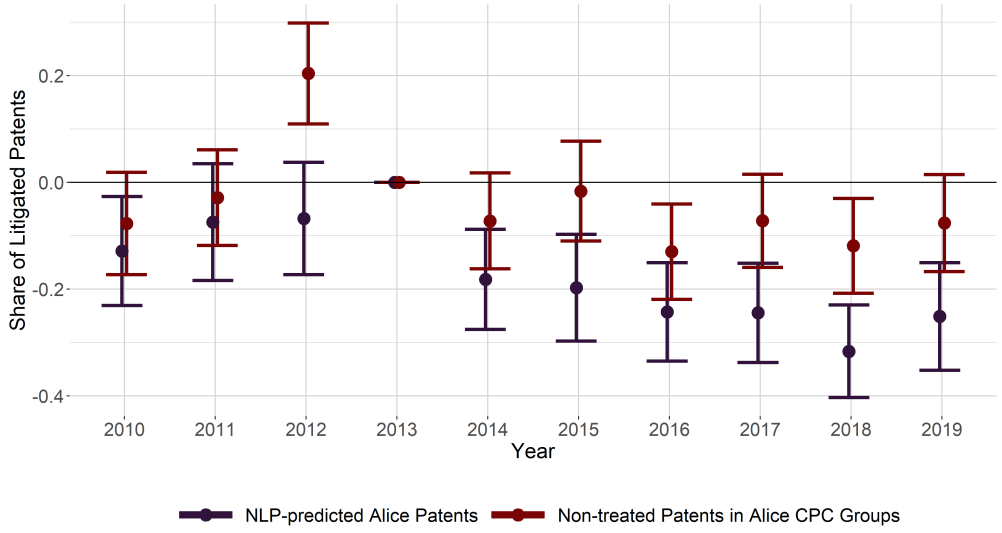
To test this prediction, I distinguish in figure 28 between the share of treated patents among the litigated patents in cases (blue), and the share of patents in the affected CPC groups that are classified as not treated (red). Thus, I compare the fractions of treated and control patents within a patent infringement case filed by a product company. I regress each of these outcomes on year dummies for the filing year of the respective lawsuit and normalize the coefficients for 2013 to zero, the year immediately before *Alice*.[77] If my NLP method can identify ineligible patents better than the CPC group alone, I should see a distinct difference in the shares of litigated patents after 2014.

While the share of non-treated patents in product company lawsuits remains relatively unchanged, we see a clear downward shift after 2014 for the share of treated patents (there is one outlier for non-treated patent shares in 2012, which is not part of a trend though). The magnitude of the shock is large; the share of treated patents before *Alice* was 4.82%, which falls to 2.49% after 2014, a drop of more than 48%. The share of non-treated patents was 16.9% and falls by 2.91 percentage points, a much smaller decrease of just 17.2%. Thus, my method can predict the enforceability of patents after *Alice*, even for patents filed within the same CPC groups. The prediction method is not perfect and there is a small treatment effect for patents that are predicted to be non-treated. However, the difference in magnitudes

[76] URL: `law.stanford.edu/projects/stanford-npe-litigation-database/`

[77] For comparability, I scale both outcomes to zero mean and unit variance. All coefficients include error bars for the 95% confidence intervals.

Figure 28 Share of treated and non-treated patents in litigation by product companies.



of the treatment effects is large, confirming that my NLP algorithm identifies the exposure to *Alice*.[78]

[78] A more detailed investigation of whether my model can predict the invalidation of patents in infringement cases is beyond the scope of this paper. To do this, case records would need to be searched in Public Access to Court Electronic Records (PACER) data and analyzed for rulings that invalidate patents and reference to the *Alice* decision. As I have shown above, greater exposure to *Alice* decreases the likelihood of being included in an infringement case since the court would likely strike down the patent as ineligible. Thus, patents that are invalidated in a court case might be less representative of affected patents in general.